

STATISTICS PAPER II · AUTHENTIC PYQS 2018-2026

Linear Models

119 authentic UPSC ISS questions with exam shortcut, tips & tricks and step-by-step solution

119

Authentic PYQs

3

Topics Covered

9

Years 2018-2026

AI-DERIVED EXPLANATIONS

The source booklets carry no official answer key. Every Exam Shortcut, Tips & Tricks and Step-by-Step Solution here is independently derived and verified by the builder of this platform, not copied from an official source.

Contents

Theory of Linear Estimation	48 questions
Estimable Functions, Estimation and Error Space	11
Gauss-Markov Theorem, Normal Equations and BLUE	19
Regression Specifics, Design Matrix, Multicollinearity	6
Variances and Covariances of LSEs; Error Variance	8
Correlated Observations, Weighted LS and GLS	4
Matrix Theory for Linear Models	34 questions
Generalized Inverse of a Matrix	22
Hat Matrix, Residuals and Quadratic Forms	12
Analysis of Variance	37 questions
Two-way Classification and ANOVA	15
One-way Classification and ANOVA	14
Multiple Comparison Tests (Tukey, Scheffe, SNK-Duncan)	4
Fixed, Random and Mixed Effects Models	4

Theory of Linear Estimation

48 authentic PYQs in this topic (2018-2026)

Estimable Functions, Estimation and Error Space

LM

2018 Q1

Conceptual

Medium

Q1.

Consider the model

$$Y_1 = \beta_1 + \beta_2 + e_1, \quad Y_2 = \beta_1 + \beta_3 + e_2, \quad Y_3 = \beta_1 + \beta_2 + e_3$$

Then $\lambda_1\beta_1 + \lambda_2\beta_2 + \lambda_3\beta_3$ is estimable if and only if

- A. $\lambda_1 + \lambda_3 = \lambda_2$
- B. $\lambda_1 + \lambda_2 = \lambda_3$
- C. $\lambda_2 + \lambda_3 = \lambda_1$
- D. $\lambda_1 + \lambda_2 + \lambda_3 = C$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Read off the rows of X : $(1,1,0)$, $(1,0,1)$, $(1,1,0)$. A function $\lambda'\beta$ is estimable iff λ' is a combination of these rows, i.e. $\lambda = (a+b, a, b)$, which forces $\lambda_1 = \lambda_2 + \lambda_3$.

TIPS & TRICKS

- Estimability test: $\lambda'\beta$ is estimable $\iff \lambda' \in \mathcal{R}(X)$, the row space of the design matrix.
- Duplicate rows (Y_3 repeats Y_1) add no new direction, so the row space here is only 2-dimensional in $\mathbb{R}^3$ - exactly one linear restriction on λ survives.
- Option (d) with an arbitrary constant C is a standard throwaway distractor: estimability is a linear-subspace condition and can never involve a free constant.

STEP-BY-STEP SOLUTION

1. Write the model as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$ with $\mathbf{X} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

2. $\lambda'\beta$ is estimable iff there is $\mathbf{c}$ with $\lambda' = \mathbf{c}'\mathbf{X}$, i.e. λ' lies in the row space of $\mathbf{X}$.

3. Rows 1 and 3 coincide, so the row space is spanned by $(1,1,0)$ and $(1,0,1)$; hence

$$\lambda = a(1,1,0) + b(1,0,1) = (a+b, a, b).$$

4. Therefore $\lambda_1 = a+b$, $\lambda_2 = a$, $\lambda_3 = b$, giving the single condition $\lambda_1 = \lambda_2 + \lambda_3$, i.e. $\lambda_2 + \lambda_3 = \lambda_1$.

5. Hence option (c) is correct.

Q2.

In the context of the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, consider the following statements:

1. If $\mathbf{X}$ is a full rank matrix, then all linear parametric functions are estimable.
2. Each estimable linear parametric function is of the form $\mathbf{c}'\mathbf{X}\boldsymbol{\beta}$ for some vector $\mathbf{c}$.

Codes:

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Both are standard theorems: full rank $\Rightarrow$ every $\lambda'\boldsymbol{\beta}$ estimable; and estimable $\iff \lambda' = \mathbf{c}'\mathbf{X}$, i.e. the function is $\mathbf{c}'\mathbf{X}\boldsymbol{\beta}$. Answer: both.

TIPS & TRICKS

- Memorise the chain: $\lambda'\boldsymbol{\beta}$ estimable $\iff \lambda' \in \mathcal{R}(\mathbf{X}) \iff \lambda' = \mathbf{c}'\mathbf{X} \iff \lambda'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \lambda'$.
- Number of linearly independent estimable functions = $\text{rank}(\mathbf{X})$.
- A common trap variant replaces ' $\text{rank}(\mathbf{X}) = p$ ' by ' $\text{rank}(\mathbf{X}) > p$ ' - impossible, since rank cannot exceed the number of columns.

STEP-BY-STEP SOLUTION

1. Statement 1: if $\mathbf{X}$ is of full column rank p , then $\mathcal{R}(\mathbf{X}) = \mathbb{R}^p$, so every $\lambda \in \mathbb{R}^p$ lies in the row space and every linear parametric function is estimable. True.
2. Statement 2: by definition $\lambda'\boldsymbol{\beta}$ is estimable iff some linear function $\mathbf{c}'\mathbf{Y}$ is unbiased for it, i.e. $E(\mathbf{c}'\mathbf{Y}) = \mathbf{c}'\mathbf{X}\boldsymbol{\beta} = \lambda'\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$. Hence every estimable function has the form $\mathbf{c}'\mathbf{X}\boldsymbol{\beta}$. True.
3. Both statements are correct - option (c).

Q3.

Consider the model $E(Y_{ij}) = \mu_i$, $\text{var}(Y_{ij}) = \sigma^2$, $j = 1, \dots, n_i$; $i = 1, \dots, p$. Which of the following is/are correct?

1. Any linear parametric function is estimable.

$$2. \hat{\mu}_i = \frac{\sum_{j=1}^{n_i} Y_{ij}}{n_i}$$

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

This is the cell-means parametrisation $E(Y_{ij}) = \mu_i$: the design matrix has p independent indicator columns, so it is full rank - every $\sum l_i \mu_i$ is estimable and $\hat{\mu}_i$ is the group mean.

TIPS & TRICKS

- Cell-means model ($E(Y_{ij}) = \mu_i$) is full rank; the over-parametrised model ($\mu + \tau_i$) is not. Read which one is printed.
- In the over-parametrised version only $\mu + \tau_i$ and contrasts $\sum c_i \tau_i$ with $\sum c_i = 0$ are estimable.
- BLUE of μ_i is always the i -th group mean $\bar{Y}_{i.}$, whatever the n_i are.

STEP-BY-STEP SOLUTION

1. Write $\mathbf{Y} = \mathbf{X}\boldsymbol{\mu} + \mathbf{e}$ where $\mathbf{X}$ has one indicator column per group. These p columns are linearly independent, so $\text{rank}(\mathbf{X}) = p$, i.e. $\mathbf{X}$ is of full column rank.
2. Full column rank implies every linear parametric function $\sum l_i \mu_i$ is estimable. Statement 1 is true.
3. The normal equations give $n_i \hat{\mu}_i = \sum_j Y_{ij}$, so $\hat{\mu}_i = \frac{\sum_j Y_{ij}}{n_i} = \bar{Y}_{i.}$. Statement 2 is true.
4. Both statements are correct - option (c).

Q4.

If y_1, y_2, y_3, y_4 are independent with $E(y_1)=E(y_2)=(\theta_1+\theta_2)$, $E(y_3)=E(y_4)=(\theta_1+\theta_3)$, $V(y_i)=\sigma^2$. For estimability of $l'\theta=l_1\theta_1+l_2\theta_2+l_3\theta_3$, consider:

1. $l_1=l_2+l_3$
2. $l_2=l_1+l_3$
3. $l_3=l_1+l_2$

Codes:

A. 1 only

B. 2 only

C. 3 only

D. 1, 2 and 3

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Rows of X are $(1,1,0)$ and $(1,0,1)$ (each twice). $l=a(1,1,0)+b(1,0,1)$ gives $l_1=l_2+l_3$: statement 1 only.

TIPS & TRICKS

- Whenever $E(y)$ repeats, the distinct rows are all that matter - duplicates add nothing to the row space.
- Estimability is a statement about l lying in $\mathcal{R}(X)$; translate the expectations into rows and read off the constraint.
- Here $\text{rank}(X)=2<3$, so exactly one linear restriction on l appears.

STEP-BY-STEP SOLUTION

1. The design matrix rows are $(1,1,0)$ for y_1, y_2 and $(1,0,1)$ for y_3, y_4 , so $\mathcal{R}(X)=\text{span}\{(1,1,0), (1,0,1)\}$.
2. $l'\theta$ is estimable iff $l=a(1,1,0)+b(1,0,1)=(a+b, a, b)$ for some a, b .
3. Then $l_1=a+b$, $l_2=a$, $l_3=b$, so the necessary and sufficient condition is $l_1=l_2+l_3$.
4. Only statement 1 gives this condition - option (a).

Q5.

For $y = X\beta + u$ ($y: n \times 1$, $X: n \times p$, $\beta: p \times 1$), $\hat{\beta}$ solves $X'X\hat{\beta} = X'y$:

1. If $C'\beta$ is estimable, then $C'\hat{\beta}$ is the BLUE.
2. All linear parametric functions are estimable iff $\text{Rank}(X) > p$.
3. $H_0: L\beta = \zeta$ may be tested using Snedecor's F statistic.

Codes:

- A. 1 only
 B. 2 only
 C. 1 and 3
 D. 2 and 3

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Statement 2 says $\text{rank}(X) > p$, which is impossible for an $n \times p$ matrix ($\text{rank} \leq p$); the condition is $\text{rank} = p$. Statements 1 and 3 are standard.

TIPS & TRICKS

- $\text{rank}(X) \leq \min(n, p)$ - any statement demanding rank strictly greater than the number of columns is automatically false.
- Gauss-Markov: for estimable $C'\beta$, $C'\hat{\beta}$ is BLUE for every solution $\hat{\beta}$ of the normal equations, and the value is invariant to which solution is used.
- A testable hypothesis $H_0: L\beta = \zeta$ (all rows of L estimable) is tested by the variance-ratio F .

STEP-BY-STEP SOLUTION

1. Statement 1: the Gauss-Markov theorem states that if $C'\beta$ is estimable then $C'\hat{\beta}$ is its BLUE. True.
2. Statement 2: every linear parametric function is estimable iff $\text{rank}(X) = p$ (full column rank). Since $\text{rank}(X) \leq p$ always, ' $\text{Rank}(X) > p$ ' is impossible. False.
3. Statement 3: an estimable linear hypothesis $H_0: L\beta = \zeta$ is tested by $F = \frac{(\text{RSS}_H - \text{RSS})/m}{\text{RSS}/(n-r)}$, Snedecor's F . True.
4. Hence 1 and 3 - option (c).

Q6.

Consider $y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$, $i=1,2$; $j=1,2$. For which choice of λ' is $\lambda'\beta$, $\beta'=(\mu, \alpha_1, \alpha_2, \beta_1, \beta_2)$, **not** estimable?

- A. (1,1,0,1,0)
- B. (0,0,0,-1,1)
- C. (0,-1,1,0,0)
- D. **✓(1,1,1,1,1)**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

In a two-way additive model, an estimable λ must satisfy (coefficient of μ) = (sum of α -coefficients) = (sum of β -coefficients). Option (d) gives 1,2,2 - not estimable.

TIPS & TRICKS

- Estimable in two-way additive layout: $\mu + \alpha_i + \beta_j$, all α -contrasts and all β -contrasts. Individual effects and sums of effects are not.
- The one-line test: write $\lambda = (\lambda_\mu, \lambda_\alpha, \lambda_\beta)$; estimability requires $\lambda_\mu = \sum \lambda_\alpha = \sum \lambda_\beta$.
- Apply the test to each option in turn: (a) 1,1,1 ok; (b) 0,0,0 ok; (c) 0,0,0 ok; (d) 1,2,2 fails.

STEP-BY-STEP SOLUTION

1. The design rows of X for $\beta'=(\mu, \alpha_1, \alpha_2, \beta_1, \beta_2)$ are (1,1,0,1,0), (1,1,0,0,1), (1,0,1,1,0), (1,0,1,0,1).
2. Any row combination $\sum c_k r_k$ has μ -coefficient $\sum c_k$, α -coefficients summing to $\sum c_k$, and β -coefficients summing to $\sum c_k$. So estimability requires $\lambda_\mu = \lambda_{\alpha_1} + \lambda_{\alpha_2} = \lambda_{\beta_1} + \lambda_{\beta_2}$.
3. (a) (1,1,0,1,0): $1=1=1$ - estimable (it is $\mu + \alpha_1 + \beta_1$, a cell mean).
4. (b) (0,0,0,-1,1): $0=0=0$ - estimable ($\beta_2 - \beta_1$, a contrast). (c) (0,-1,1,0,0): $0=0=0$ - estimable ($\alpha_2 - \alpha_1$).
5. (d) (1,1,1,1,1): $\lambda_\mu=1$ but $\sum \lambda_\alpha=2$ - the condition fails, so $\mu + \alpha_1 + \alpha_2 + \beta_1 + \beta_2$ is NOT estimable. Option (d).

Q7.

For $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, $i=1,2; j=1,2$:

1. $\mu + \tau_1$ is estimable.
2. $\tau_1 + \tau_2$ is estimable.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT AI-derived explanation

In $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ the estimable functions are $\mu + \tau_i$ and contrasts $\sum c_i \tau_i$ with $\sum c_i = 0$. $\tau_1 + \tau_2$ has $\sum c_i = 2 \neq 0$.

TIPS & TRICKS

- Rule for one-way over-parametrised models: estimable $\iff \lambda_\mu = \sum \lambda_{\tau_i}$.
- $\mu + \tau_i$ is the i -th cell mean, estimated by $\bar{y}_{i.}$ - always estimable.
- $\tau_1 - \tau_2$ is estimable but $\tau_1 + \tau_2$ and τ_1 alone are not - the difference between a contrast and a sum is the whole point.

STEP-BY-STEP SOLUTION

1. Design rows are $(1,1,0)$ (for $i=1$) and $(1,0,1)$ (for $i=2$) with $\beta' = (\mu, \tau_1, \tau_2)$.
2. A combination $a(1,1,0) + b(1,0,1) = (a+b, a, b)$, so estimability requires $\lambda_\mu = \lambda_{\tau_1} + \lambda_{\tau_2}$.
3. $\mu + \tau_1$ is $(1,1,0)$: $1 = 1 + 0$ - estimable, with BLUE $\bar{y}_{1.}$.. Statement 1 true.
4. $\tau_1 + \tau_2$ is $(0,1,1)$: $\lambda_\mu = 0$ but $\lambda_{\tau_1} + \lambda_{\tau_2} = 2$ - not estimable. Statement 2 false.
5. Hence 1 only - option (a).

Q8.

Two-way classification, one observation per cell: $y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}$; $i = 1, \dots, p$; $j = 1, \dots, q$. Which parameter is estimable?

- A. α
- B. β
- C. $\alpha_i - \alpha_u$; $i, u = 1, \dots, p$ ($i \neq u$)
- D. $\alpha_1 + \alpha_2$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

In the additive two-way layout only $\mu + \alpha_i + \beta_j$ and contrasts of the α 's or β 's are estimable. That is option (c), $\alpha_i - \alpha_u$.

TIPS & TRICKS

- Estimability test: $\lambda_\mu = \sum \lambda_\alpha = \sum \lambda_\beta$. For $\alpha_i - \alpha_u$ all three are zero, so it passes.
- $\alpha_1 + \alpha_2$ has $\sum \lambda_\alpha = 2 \neq 0 = \lambda_\mu$ - fails. Individual α_i or β_j likewise fail.
- General principle in over-parametrised models: differences are estimable, levels are not.

STEP-BY-STEP SOLUTION

1. Rows of the design matrix are $(1, \mathbf{e}_i, \mathbf{e}_j)$, so any estimable λ satisfies $\lambda_\mu = \sum_i \lambda_{\alpha_i} = \sum_j \lambda_{\beta_j}$.
2. α alone and β alone are not even well-defined single parameters, and individual effects fail the condition.
3. $\alpha_1 + \alpha_2$: $\lambda_\mu = 0$ but $\sum \lambda_\alpha = 2$ - not estimable.
4. $\alpha_i - \alpha_u$: $\lambda_\mu = 0$, $\sum \lambda_\alpha = 1 - 1 = 0$, $\sum \lambda_\beta = 0$ - estimable, with BLUE $\bar{y}_{i\cdot} - \bar{y}_{u\cdot}$. Option (c).

Q9.

Consider the model $E(Y_{ij}) = \alpha_i + \beta_j$; $i=1,2$ and $j=1,2$, where Y_{ij} are uncorrelated with common variance σ^2 . Then the function $l_1\alpha_1 + l_2\alpha_2 + m_1\beta_1 + m_2\beta_2$ is estimable if and only if:

- A. $l_1 - l_2 = m_1 - m_2$
- B. $l_1 + l_2 = m_1 + m_2$
- C. $l_1 + l_2 + m_1 + m_2 = 0$
- D. $l_1 + l_2 + m_1 + m_2 = 1$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Rows are (1,0,1,0), (1,0,0,1), (0,1,1,0), (0,1,0,1): any combination has $\sum l = \sum m$, so the condition is $l_1 + l_2 = m_1 + m_2$.

TIPS & TRICKS

- Two-way additive layout without a general mean: the α -coefficients and the β -coefficients must have equal totals.
- $\text{rank}(X) = 2 + 2 - 1 = 3 < 4$, so exactly one linear restriction on (l_1, l_2, m_1, m_2) appears.
- Check any candidate against a cell mean: $\alpha_1 + \beta_1$ has l -sum 1 and m -sum 1 - consistent with option (b).

STEP-BY-STEP SOLUTION

- With parameter vector $(\alpha_1, \alpha_2, \beta_1, \beta_2)$ the four design rows are (1,0,1,0), (1,0,0,1), (0,1,1,0), (0,1,0,1).
- A general row combination $\sum_k c_k r_k$ has $l_1 = c_1 + c_2$, $l_2 = c_3 + c_4$, $m_1 = c_1 + c_3$, $m_2 = c_2 + c_4$.
- Hence $l_1 + l_2 = \sum_k c_k = m_1 + m_2$ for every estimable function, and conversely this single condition is sufficient.
- Therefore the function is estimable iff $l_1 + l_2 = m_1 + m_2$ - option (b).

Q10.

Let $\lambda'\beta$ be an estimable function of β in the model $Y = X\beta + \varepsilon$, where $E(Y) = X\beta$ and X is an $n \times p$ matrix of rank $k < p \leq n$. Let $\hat{\beta}$ be any solution to the normal equations $X'X\hat{\beta} = X'Y$ and let r be any solution to $X'Xr = \lambda$. Consider the following statements for the two estimators $\lambda'\hat{\beta}$ and $r'X'Y$:

- I. $\lambda'\hat{\beta}$ is equal to $r'X'Y$ for any $\hat{\beta}$ or r .
- II. $\lambda'\hat{\beta}$ and $r'X'Y$ are invariant to the choice of any $\hat{\beta}$ or r .

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$r'X'Y = (X'Xr)'\hat{\beta} = \lambda'\hat{\beta}$ using the normal equations, so the two estimators coincide and both are invariant.

TIPS & TRICKS

- Key manipulation: $r'X'Y = r'(X'X\hat{\beta}) = (X'Xr)'\hat{\beta} = \lambda'\hat{\beta}$.
- For an estimable $\lambda'\beta$, the value $\lambda'\hat{\beta}$ does not depend on which solution $\hat{\beta}$ of the normal equations is used - that is the substance of the Gauss-Markov theorem in the less-than-full-rank case.
- Estimability is essential: for a non-estimable λ , $\lambda'\hat{\beta}$ would change with $\hat{\beta}$ and $X'Xr = \lambda$ would have no solution.

STEP-BY-STEP SOLUTION

1. $\hat{\beta}$ satisfies $X'X\hat{\beta} = X'Y$ and r satisfies $X'Xr = \lambda$.
2. Then $r'X'Y = r'(X'X\hat{\beta}) = (X'Xr)'\hat{\beta} = \lambda'\hat{\beta}$, for every admissible choice of $\hat{\beta}$ and r . Statement I is true.
3. For estimable $\lambda'\beta$ write $\lambda' = c'X$; then $\lambda'\hat{\beta} = c'X\hat{\beta} = c'PY$, which depends only on the projection PY and not on the particular solution. Statement II is true.
4. Both correct - option (c).

Q11.

In the Gauss-Markov linear model $\mathbf{Y}_{n \times 1} = \mathbf{X}_{n \times k} \boldsymbol{\beta}_{k \times 1} + \boldsymbol{\varepsilon}_{n \times 1}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ with $\text{rank } r(\mathbf{X}) = r < k$, the set $\{\mathbf{l} \in \mathbb{R}^n : E(\mathbf{l}'\mathbf{Y}) = 0\}$ is

- A. not a subspace
- B. a subspace of $\mathbb{R}^n$ with dimension $k-r$
- C. a subspace of $\mathbb{R}^n$ with dimension $n-k$
- D. a subspace of $\mathbb{R}^n$ with dimension $n-r$**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$E(\mathbf{l}'\mathbf{Y}) = \mathbf{l}'\mathbf{X}\boldsymbol{\beta} = 0$ for all $\boldsymbol{\beta}$ means $\mathbf{X}'\mathbf{l} = \mathbf{0}$, i.e. $\mathbf{l} \in \mathcal{C}(\mathbf{X})^\perp$ - the error space, of dimension $n-r$.

TIPS & TRICKS

- Estimation space = $\mathcal{C}(\mathbf{X})$, dimension r . Error space = $\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}')$, dimension $n-r$. They partition $\mathbb{R}^n$.
- The set is the null space of a linear map, hence automatically a subspace - option (a) can be discarded at once.
- $n-r$ is exactly the residual degrees of freedom, which is the practical meaning of this dimension.

STEP-BY-STEP SOLUTION

1. $E(\mathbf{l}'\mathbf{Y}) = \mathbf{l}'\mathbf{X}\boldsymbol{\beta}$. This vanishes identically in $\boldsymbol{\beta}$ if and only if $\mathbf{l}'\mathbf{X} = \mathbf{0}'$, i.e. $\mathbf{X}'\mathbf{l} = \mathbf{0}$.
2. So the set is $\mathcal{N}(\mathbf{X}')$, the null space of $\mathbf{X}'$ - a linear subspace of $\mathbb{R}^n$ (it is closed under addition and scalar multiplication).
3. By rank-nullity applied to $\mathbf{X}'$ (an $k \times n$ matrix of rank r), $\dim \mathcal{N}(\mathbf{X}') = n-r$.
4. This is the error space, the orthogonal complement of the estimation space $\mathcal{C}(\mathbf{X})$. Hence option (d).

Gauss-Markov Theorem, Normal Equations and BLUE

Q1.

Consider the model $y_i = \beta_0 + \beta_1 x_i + e_i$ ($i=1,2,3$) where $x_1=-1$, $x_2=0$, $x_3=1$. The best linear unbiased estimate of β_0 and β_1 is

where $\bar{y} = \frac{y_1 + y_2 + y_3}{3}$.

A. $\left(\bar{y}, \frac{y_3 - y_1}{2}\right)$

B. $\left(\frac{y_2 - y_1}{2}, \frac{y_3 - y_1}{2}\right)$

C. $\left(\bar{y}, \frac{y_2 - y_1}{2}\right)$

D. $\left(\frac{y_1 + y_2}{2}, \bar{y}\right)$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

The x 's are symmetric about 0, so $\bar{x}=0$: the design is orthogonal and $\hat{\beta}_0 = \bar{y}$, $\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{y_3 - y_1}{2}$.

TIPS & TRICKS

- When $\sum x_i = 0$ the two normal equations decouple - no need to invert a 2×2 matrix.
- $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$; here $S_{xy} = (-1)y_1 + 0 \cdot y_2 + (1)y_3$ and $S_{xx} = 2$.
- Spot-check: any option whose intercept is not $\bar{y}$ can be eliminated instantly, because $\bar{x}=0$ forces $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \bar{y}$.

STEP-BY-STEP SOLUTION

1. Here $x_1=-1$, $x_2=0$, $x_3=1$, so $\bar{x}=0$ and $S_{xx} = \sum (x_i - \bar{x})^2 = 1 + 0 + 1 = 2$.

2. $S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = (-1)y_1 + 0 \cdot y_2 + (1)y_3 = y_3 - y_1$.

3. Hence $\hat{\beta}_1 = S_{xy}/S_{xx} = \frac{y_3 - y_1}{2}$.

4. And $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \bar{y}$ because $\bar{x}=0$.

5. So the BLUE pair is $(\bar{y}, \frac{y_3 - y_1}{2})$ - option (a). By Gauss-Markov these least-squares estimators are BLUE.

LM

2019 Q25

Conceptual

Easy

Q2.

For $y = X\beta + u$; $E(u) = 0$, $V(u) = \sigma^2 I$, $\hat{\beta} = (X'X)^{-1}X'y$:

1. $V(\hat{\beta}) = \sigma^2(X'X)^{-1}$.
2. $\hat{\beta}$ is BLUE for β .
3. Residual mean square is an unbiased estimator of σ^2 .

Codes:

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

All three are the standard Gauss-Markov package: $V(\hat{\beta}) = \sigma^2(X'X)^{-1}$, $\hat{\beta}$ is BLUE, and $MSE = RSS/(n-p)$ is unbiased for σ^2 .

TIPS & TRICKS

- Gauss-Markov needs only $E(u) = 0$ and $V(u) = \sigma^2 I$ - normality is not required for BLUE.
- $E(RSS) = (n-r)\sigma^2$ with $r = \text{rank}(X)$, hence the divisor $n-r$ (here $n-p$).
- Do not confuse 'BLUE' (best among linear unbiased) with 'MVUE' (best among all unbiased) - the latter needs normality.

STEP-BY-STEP SOLUTION

1. $\hat{\beta} = (X'X)^{-1}X'y$, so $V(\hat{\beta}) = (X'X)^{-1}X' \sigma^2 I X(X'X)^{-1} = \sigma^2(X'X)^{-1}$. Statement 1 true.
2. By the Gauss-Markov theorem, under $E(u) = 0$, $V(u) = \sigma^2 I$, the least-squares estimator is the best linear unbiased estimator. Statement 2 true.
3. $E(RSS) = \sigma^2(n-p)$, so the residual mean square $RSS/(n-p)$ is unbiased for σ^2 . Statement 3 true.
4. All three correct - option (d).

Q3.

Consider $E(y_1)=2\beta_1+\beta_2$, $E(y_2)=\beta_1-\beta_2$, $E(y_3)=\beta_1+\alpha\beta_2$. The value of α such that the BLUEs of β_1, β_2 are uncorrelated is

A. -1**B. +1****C. 0****D. +2****Correct answer: A****EXAM SHORTCUT** AI-derived explanation

BLUEs are uncorrelated iff the off-diagonal of $X'X$ vanishes. Here it is $2(1)+1(-1)+1(\alpha)=1+\alpha$, so $\alpha=-1$.

TIPS & TRICKS

- $\text{Cov}(\hat{\beta}) = \sigma^2(X'X)^{-1}$; for a 2×2 matrix the inverse has zero off-diagonal exactly when the original does.
- Equivalently, the two columns of the design matrix must be orthogonal - this reduces the whole question to one dot product.
- This exact item reappeared as 2025 Q59 with the same options, and as 2026 Q55 with new coefficients. Own the dot-product method.

STEP-BY-STEP SOLUTION

1. The design matrix is $X = \begin{pmatrix} 2 & 1 \\ 1 & -1 \\ 1 & \alpha \end{pmatrix}$ with columns $c_1 = (2, 1, 1)'$ and $c_2 = (1, -1, \alpha)'$.
2. $X'X = \begin{pmatrix} c_1'c_1 & c_1'c_2 \\ c_1'c_2 & c_2'c_2 \end{pmatrix}$ and $\text{Cov}(\hat{\beta}) = \sigma^2(X'X)^{-1}$.
3. For a 2×2 symmetric matrix, $(X'X)^{-1}$ has zero off-diagonal iff $X'X$ does, i.e. iff $c_1'c_2 = 0$.
4. $c_1'c_2 = 2(1) + 1(-1) + 1(\alpha) = 1 + \alpha = 0 \Rightarrow \alpha = -1$.
5. Hence option (a).

Q4.

Let $X_i, Y_i, Z_i; i=1,2,3$ be nine independent observations with common variance σ^2 , $E(X_i)=\theta_1$, $E(Y_i)=\theta_2$, $E(Z_i)=\theta_1-\theta_2$. If $X=\sum_{i=1}^3 x_i$, $Y=\sum_{i=1}^3 y_i$, $Z=\sum_{i=1}^3 z_i$, the BLUE of θ_1 is given by

- A. $\frac{\sqrt{1}}{9}[2X+Y-Z]$
- B. $\frac{1}{9}[X+2Y-Z]$
- C. $\frac{1}{9}[X+2Y-2Z]$
- D. $\frac{1}{9}[X+3Y-2Z]$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Build $X'X=3\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, invert to $\frac{1}{9}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and apply to $X'Y=(X+Z, Y-Z)'$.

TIPS & TRICKS

- The nine-observation family has been asked in 2021, 2022, 2023 and 2025 with permuted signs. Learn the drill, not the answer.
- Always audit a candidate BLUE by unbiasedness: substitute $E(X)=3\theta_1$, $E(Y)=3\theta_2$, $E(Z)=3(\theta_1-\theta_2)$ and check the result reduces to θ_1 .
- $\frac{1}{9}[2X+Y+Z]$ passes that audit; $\frac{1}{9}[X+2Y-Z]$ is the BLUE of θ_2 , a deliberate near-miss.

STEP-BY-STEP SOLUTION

1. Design rows: (1,0) three times for the X_i , (0,1) three times for the Y_i , (1,-1) three times for the Z_i .

2. $X'X=3\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}+3\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}+3\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}=\begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$, with $(X'X)^{-1}=\frac{1}{9}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

3. $X'Y=\begin{pmatrix} X+Z \\ Y-Z \end{pmatrix}$, so $\hat{\theta}_1=\frac{1}{9}[2(X+Z)+(Y-Z)]=\frac{1}{9}[2X+Y+Z]$ and $\hat{\theta}_2=\frac{1}{9}[X+2Y-Z]$.

4. Unbiasedness check: $E\left(\frac{1}{9}[2X+Y+Z]\right)=\frac{1}{9}[6\theta_1+3\theta_2+3\theta_1-3\theta_2]=\theta_1$.

5. The intended key is option (a), which is this expression up to the printed sign of Z (see the source flag).

Source note: Source anomaly: under the printed model $E(Z_i)=\theta_1-\theta_2$ the exact BLUE is $\frac{1}{9}[2X+Y+Z]$. The identical item in 2023 (Q56) carries $\frac{1}{9}[2X+Y+Z]$ among its options and that is its key. Option (a) here differs only in the sign of Z , indicating a sign typo in the printed option. Option (a) is taken as the intended key; the source wording and options are preserved unchanged.

Q5.

Consider the model

$$Y_1 = A + B + C + D + e_1; \quad Y_2 = A + C - B - D + e_2; \quad Y_3 = A + B - C - D + e_3; \quad Y_4 = A + D - B - C + e_4$$

If this is equivalent to $Y = X\beta + e$, then $(X'X)^{-1}$ equals

A. $\frac{1}{3} I_4$

B. $\frac{1}{2} I_4$

C. $\frac{1}{5} I_4$

D. $\frac{1}{4} I_4$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

The four rows are ± 1 patterns forming an orthogonal (Hadamard-type) design, so $X'X = 4I_4$ and

$$(X'X)^{-1} = \frac{1}{4} I_4.$$

TIPS & TRICKS

- Each column has four entries of ± 1 , so every diagonal entry of $X'X$ is 4; orthogonality kills the off-diagonals.
- Orthogonal design $\Rightarrow X'X$ diagonal $\Rightarrow$ uncorrelated BLUEs, each with variance $\sigma^2/4$.
- You never need to invert anything: for a design with ± 1 entries and n rows, $X'X = nI$ when columns are orthogonal.

STEP-BY-STEP SOLUTION

1. With $\beta' = (A, B, C, D)$ the design matrix is $X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$

2. Every column has entries ± 1 , so each column has squared norm 4: the diagonal of $X'X$ is (4,4,4,4).

3. Any two distinct columns agree in exactly two positions and differ in two, so their inner product is 0: the off-diagonals vanish.

4. Hence $X'X=4I_4$ and $(X'X)^{-1}=\frac{1}{4}I_4$ - option (d).

Q6.

An ecologist takes data $(X_i, Y_i); i = 1, 2, 3, \dots, n$ where X_i is the size of an area and Y_i is the number of moss plants in the area. We model the data by $Y_i \sim \text{Poisson}(\theta x_i)$; Y_i 's are independent. The least square estimator and the MLE of θ are respectively:

A. $\left(\frac{\sum_{i=1}^n Y_i}{\sum_{i=1}^n X_i}, \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2} \right)$

B. ✓ $\left(\frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}, \frac{\sum_{i=1}^n Y_i}{\sum_{i=1}^n X_i} \right)$

C. $\left(\frac{\bar{Y}}{\bar{X}}, \frac{\text{Cov}(X, Y)}{V(X)} \right)$

D. $\left(\frac{\text{Cov}(X, Y)}{V(X)}, \frac{\bar{Y}}{\bar{X}} \right)$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Least squares minimises $\sum (Y_i - \theta x_i)^2$ giving $\sum x_i Y_i / \sum x_i^2$; the Poisson likelihood gives $\sum Y_i / \sum x_i$. In that order, option (b).

TIPS & TRICKS

- Least squares ignores the variance structure; the Poisson MLE automatically weights by the mean, so the two differ here.
- Poisson log-likelihood: $\ell = -\theta \sum x_i + \sum Y_i \log \theta + \text{const}$, so $\hat{\theta}_{MLE} = \sum Y_i / \sum x_i$.
- Read the order in which the question asks - 'least square estimator and the MLE respectively' - options (a) and (b) are the same pair swapped.

STEP-BY-STEP SOLUTION

1. Least squares: minimise $S(\theta) = \sum_i (Y_i - \theta x_i)^2$. Then $S'(\theta) = -2 \sum x_i (Y_i - \theta x_i) = 0$ gives $\hat{\theta}_{LS} = \frac{\sum x_i Y_i}{\sum x_i^2}$.

2. Maximum likelihood: $L(\theta) = \prod_i e^{-\theta x_i} \frac{(\theta x_i)^{Y_i}}{Y_i!}$, so $\ell(\theta) = -\theta \sum x_i + \left(\sum Y_i \right) \log \theta + \text{const}$.

3. $\frac{\partial \ell}{\partial \theta} = -\sum x_i + \frac{\sum Y_i}{\theta} = 0 \Rightarrow \hat{\theta}_{MLE} = \frac{\sum Y_i}{\sum x_i}$.

4. So (LS, MLE) = $\left(\frac{\sum X_i Y_i}{\sum X_i^2}, \frac{\sum Y_i}{\sum X_i} \right)$ - option (b).

LM

2022 Q6

Conceptual

Medium

Q7.

Let X_i , Y_i and Z_i ; $i = 1, 2, 3$ be nine independent observations with common variance σ^2 , and $E(X_i) = \theta_1$, $E(Y_i) = \theta_2$, $E(Z_i) = \theta_1 - \theta_2$; $i = 1, 2, 3$. If $\hat{\theta}_1$ and $\hat{\theta}_2$ denote the BLUEs of θ_1 and θ_2 respectively, then which one of the following is correct?

A. $V(\hat{\theta}_1) > V(\hat{\theta}_2)$

B. $V(\hat{\theta}_1) = V(\hat{\theta}_2)$

C. $V(\hat{\theta}_1) < V(\hat{\theta}_2)$

D. $V(\hat{\theta}_1) \geq V(\hat{\theta}_2)$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$X'X = 3 \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ has equal diagonal entries, so $(X'X)^{-1} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $V(\hat{\theta}_1) = V(\hat{\theta}_2) = \frac{2\sigma^2}{9}$.

TIPS & TRICKS

- You never need the estimators themselves: symmetry of the design in θ_1 and θ_2 already forces equal variances.
- $V(\hat{\theta}) = \sigma^2(X'X)^{-1}$; read the variances off the diagonal.
- Whenever the roles of two parameters can be swapped by relabelling the observations, their BLUEs must have equal variance - a 5-second argument.

STEP-BY-STEP SOLUTION

1. Design rows: (1,0) three times, (0,1) three times, (1,-1) three times.

2. $X'X = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$, so $(X'X)^{-1} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

3. $V(\hat{\theta}_1) = \sigma^2 \cdot \frac{2}{9}$ and $V(\hat{\theta}_2) = \sigma^2 \cdot \frac{2}{9}$.

4. They are equal - option (b).

SHARED CONTEXT

Let X_i , Y_i and Z_i where $i = 1, 2, 3, 4$ be twelve independent observations with common variance σ^2 , and $E(X_i) = \theta_1$, $E(Y_i) = \theta_2$ and $E(Z_i) = \theta_1 + \theta_2$; $i = 1, 2, 3, 4$. Let $X = \sum_{i=1}^4 X_i$, $Y = \sum_{i=1}^4 Y_i$ and $Z = \sum_{i=1}^4 Z_i$.

Q8.

What is the BLUE of θ_1 equal to?

A. $\frac{1}{12}[2X - Y + Z]$

B. $\frac{1}{12}[X - Y + 2Z]$

C. $\frac{1}{12}[-X + 2Y + Z]$

D. $\frac{1}{12}[X - 2Y + Z]$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$$X'X = \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}, (X'X)^{-1} = \frac{1}{12} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, X'Y = (X+Z, Y+Z)', \text{ giving } \hat{\theta}_1 = \frac{1}{12}[2X - Y + Z].$$

TIPS & TRICKS

- With $E(Z_i) = \theta_1 + \theta_2$ the off-diagonal of $X'X$ is +4, so the inverse carries -1 off-diagonal - the sign flip is where marks are lost.
- Audit by unbiasedness: $E\left(\frac{1}{12}[2X - Y + Z]\right) = \frac{1}{12}[8\theta_1 - 4\theta_2 + 4\theta_1 + 4\theta_2] = \theta_1$.
- Same skeleton as 2021 Q3, 2023 Q56-57 and 2025 Q61-62 - only the signs in $E(Z_i)$ move.

STEP-BY-STEP SOLUTION

1. Design rows: (1,0) four times, (0,1) four times, (1,1) four times.

$$2. X'X = 4 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 4 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + 4 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix}, |X'X| = 48.$$

$$3. (X'X)^{-1} = \frac{1}{48} \begin{pmatrix} 8 & -4 \\ -4 & 8 \end{pmatrix} = \frac{1}{12} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \text{ and } X'Y = \begin{pmatrix} X+Z \\ Y+Z \end{pmatrix}.$$

$$4. \hat{\theta}_1 = \frac{1}{12} [2(X+Z) - (Y+Z)] = \frac{1}{12} [2X - Y + Z].$$

5. Hence option (a).

SHARED CONTEXT

Let X_i , Y_i and Z_i where $i = 1, 2, 3, 4$ be twelve independent observations with common variance σ^2 , and $E(X_i) = \theta_1$, $E(Y_i) = \theta_2$ and $E(Z_i) = \theta_1 + \theta_2$; $i = 1, 2, 3, 4$. Let $X = \sum_{i=1}^4 X_i$, $Y = \sum_{i=1}^4 Y_i$ and $Z = \sum_{i=1}^4 Z_i$.

Q9.

What is the BLUE of θ_2 equal to?

- A. $\frac{1}{12}[2X - Y + Z]$
- B. $\frac{1}{12}[X - Y + 2Z]$
- C. $\frac{1}{12}[-X + 2Y + Z]$
- D. $\frac{1}{12}[X - 2Y + Z]$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Second row of $(X'X)^{-1}X'Y$: $\hat{\theta}_2 = \frac{1}{12}[-(X+Z)+2(Y+Z)] = \frac{1}{12}[-X+2Y+Z]$.

TIPS & TRICKS

- By the symmetry of the design in θ_1, θ_2 , the answer to this item is the previous one with X and Y interchanged.
- Unbiasedness audit: $\frac{1}{12}[-4\theta_1 + 8\theta_2 + 4\theta_1 + 4\theta_2] = \theta_2$.
- Both BLUEs have variance $\frac{2\sigma^2}{12} = \frac{\sigma^2}{6}$ since the diagonal of $(X'X)^{-1}$ is $(2/12, 2/12)$.

STEP-BY-STEP SOLUTION

1. With $(X'X)^{-1} = \frac{1}{12} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ and $X'Y = (X+Z, Y+Z)'$:

2. $\hat{\theta}_2 = \frac{1}{12}[-(X+Z)+2(Y+Z)]$.

3. $= \frac{1}{12}[-X+2Y+Z]$.

4. Hence option (c).

SHARED CONTEXT

Consider the Gauss-Markov linear model

$$E(y_1) = \beta_1 - \beta_3, \quad E(y_2) = 2\beta_1, \quad E(y_3) = \beta_1 + \beta_2$$

with $V(y_i) = \sigma^2$, $\text{Cov}(y_i, y_j) = 0$ for $i \neq j$.

Q10.

Consider the following statements:

1. The best linear unbiased estimator of β_1 is $\hat{\beta}_1 = \frac{1}{6}(y_1 + 2y_2 + y_3)$.
2. The best linear unbiased estimator of β_2 is $\hat{\beta}_2 = \frac{1}{2}(y_1 + y_3)$.

Which of the statements given above is/are correct?

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

With $X = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}$, $X'X = \text{diag}(6, 2)$, so $\hat{\beta}_1 = \frac{1}{6}(y_1 + 2y_2 + y_3)$ and $\hat{\beta}_2 = \frac{1}{2}(y_3 - y_1)$ - statement 1 true, statement 2 has the wrong sign.

TIPS & TRICKS

- An orthogonal design ($X'X$ diagonal) lets you read each BLUE off separately: $\hat{\beta}_j = \frac{\text{column}_j'y}{|\text{column}_j|^2}$.
- Always unbiasedness-test a printed candidate: $E\left[\frac{1}{2}(y_1 + y_3)\right] = \beta_1$, not β_2 , so statement 2 falls at once.
- $\frac{1}{2}(y_3 - y_1)$ versus $\frac{1}{2}(y_1 + y_3)$ - sign traps of this kind are the examiner's favourite device in this cluster.

STEP-BY-STEP SOLUTION

1. Take $\beta' = (\beta_1, \beta_2)$ with $X = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}$; the two columns are orthogonal since $1(-1) + 2(0) + 1(1) = 0$.

2. $X'X = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$ and $X'y = \begin{pmatrix} y_1 + 2y_2 + y_3 \\ -y_1 + y_3 \end{pmatrix}$.

3. $\hat{\beta}_1 = \frac{y_1 + 2y_2 + y_3}{6}$ - statement 1 is correct.

4. $\hat{\beta}_2 = \frac{y_3 - y_1}{2}$, not $\frac{y_1 + y_3}{2}$ - statement 2 is incorrect.

5. Hence 1 only - option (a).

Source note: Source anomaly: the printed stem writes $E(y_1) = \beta_1 - \beta_3$. With three free parameters and only three observations the model would be saturated (RSS = 0), and neither statement nor any option in the companion item could hold. Under the standard two-parameter form $E(y_1) = \beta_1 - \beta_2$ the arithmetic matches the printed options exactly, so the key is set on that reading. Source wording and options are preserved unchanged.

SHARED CONTEXT

Let X_i, Y_i, Z_i where $i=1,2,3$ be nine independent observations with common variance σ^2 , and $E(X_i)=\theta_1$, $E(Y_i)=\theta_2$, $E(Z_i)=\theta_1-\theta_2$; $i=1,2,3$. Let X, Y, Z respectively denote $\sum_{i=1}^3 X_i$, $\sum_{i=1}^3 Y_i$, $\sum_{i=1}^3 Z_i$.

Q11.

What is the BLUE of θ_1 equal to?

A. $\frac{1}{9}[2X+3Y-Z]$

B. $\frac{1}{9}[3X+Y-2Z]$

C. $\frac{1}{9}[2X+Y+Z]$

D. $\frac{1}{9}[2X+Y-Z]$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$X'X = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}, (X'X)^{-1} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, X'Y = (X+Z, Y-Z)', \text{ so } \hat{\theta}_1 = \frac{1}{9}[2X+Y+Z].$$

TIPS & TRICKS

- With $E(Z_i)=\theta_1-\theta_2$ the off-diagonal of $X'X$ is -3 , so the inverse has $+1$ off-diagonal - hence the $+Z$ that many candidates miss.
- Audit: $E\left(\frac{1}{9}[2X+Y+Z]\right) = \frac{1}{9}[6\theta_1+3\theta_2+3\theta_1-3\theta_2] = \theta_1$.
- $V(\hat{\theta}_1) = \frac{2\sigma^2}{9}$, the $(1,1)$ element of $\sigma^2(X'X)^{-1}$.

STEP-BY-STEP SOLUTION

1. Design rows: $(1,0)$ three times, $(0,1)$ three times, $(1,-1)$ three times.

2. $X'X = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$ with $|X'X|=27$, so $(X'X)^{-1} = \frac{1}{27} \begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

3. $X'Y = \begin{pmatrix} X+Z \\ Y-Z \end{pmatrix}$.

4. $\hat{\theta}_1 = \frac{1}{9}[2(X+Z)+(Y-Z)] = \frac{1}{9}[2X+Y+Z]$.

5. Hence option (c).

SHARED CONTEXT

Let X_i, Y_i, Z_i where $i=1,2,3$ be nine independent observations with common variance σ^2 , and $E(X_i)=\theta_1$, $E(Y_i)=\theta_2$, $E(Z_i)=\theta_1-\theta_2$; $i=1,2,3$. Let X, Y, Z respectively denote $\sum_{i=1}^3 X_i$, $\sum_{i=1}^3 Y_i$, $\sum_{i=1}^3 Z_i$.

Q12.

What is the BLUE of θ_2 equal to?

- A. $\frac{1}{9}[X+3Y-2Z]$
- B. $\frac{1}{9}[3X+Y-2Z]$
- C. $\frac{1}{9}[2X+Y-Z]$
- D. $\frac{1}{9}[X+2Y-Z]$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Second component: $\hat{\theta}_2 = \frac{1}{9}[(X+Z)+2(Y-Z)] = \frac{1}{9}[X+2Y-Z]$.

TIPS & TRICKS

- Swap the rows of $(X'X)^{-1}$ - the second row (1,2) acts on the same $X'Y$ vector.
- Audit: $E\left(\frac{1}{9}[X+2Y-Z]\right) = \frac{1}{9}[3\theta_1+6\theta_2-3\theta_1+3\theta_2] = \theta_2$.
- Note $\frac{1}{9}[2X+Y-Z]$ appears as a distractor in both parts of this pair - it is unbiased for neither parameter.

STEP-BY-STEP SOLUTION

1. Using $(X'X)^{-1} = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $X'Y = (X+Z, Y-Z)'$:

2. $\hat{\theta}_2 = \frac{1}{9}[1 \cdot (X+Z) + 2 \cdot (Y-Z)]$.

3. $= \frac{1}{9}[X+2Y-Z]$.

4. Hence option (d).

Q13.

For a linear model, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $D(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ where D stands for variance-covariance matrix and $\mathbf{I}$, for the identity matrix of order n ,

I. The observational equation $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ is always consistent.

II. The normal equation $\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{Y}$ always admits a solution.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ need not be consistent (the data rarely lie exactly in $\mathcal{C}(\mathbf{X})$), but $\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{X}'\mathbf{Y}$ is always consistent since $\mathcal{C}(\mathbf{X}'\mathbf{Y}) \subseteq \mathcal{C}(\mathbf{X}'\mathbf{X})$.

TIPS & TRICKS

- The whole point of least squares is that $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ is generally inconsistent - otherwise there would be no residuals.
- $\mathcal{C}(\mathbf{X}') = \mathcal{C}(\mathbf{X}'\mathbf{X})$, so $\mathbf{X}'\mathbf{Y}$ always lies in the column space of $\mathbf{X}'\mathbf{X}$: the normal equations always have a solution.
- When $\text{rank}(\mathbf{X}) < p$ the solution is not unique, but it always exists - existence and uniqueness are different questions.

STEP-BY-STEP SOLUTION

1. Statement I: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ is solvable only if $\mathbf{Y} \in \mathcal{C}(\mathbf{X})$, which fails whenever there is genuine error. Not always consistent - false.
2. Statement II: $\mathcal{C}(\mathbf{X}'\mathbf{X}) = \mathcal{C}(\mathbf{X}')$, and $\mathbf{X}'\mathbf{Y} \in \mathcal{C}(\mathbf{X}')$, so $\mathbf{X}'\mathbf{Y} \in \mathcal{C}(\mathbf{X}'\mathbf{X})$ and the normal equations are always consistent. True.
3. Hence II only - option (b).

Q14.

Let the model be defined as

$$E(Y_1) = 2\beta_1 + \beta_2, \quad E(Y_2) = \beta_1 - \beta_2, \quad E(Y_3) = \beta_1 + \alpha\beta_2$$

where Y_i 's are uncorrelated and have a constant variance σ^2 . What is the value of α so that the least square estimates of β_1 and β_2 are uncorrelated?

- A. 2
- B. 1
- C. -1
- D. -2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Columns $(2,1,1)'$ and $(1,-1,\alpha)'$ must be orthogonal: $2 \cdot 1 + \alpha = 0 \Rightarrow \alpha = -1$.

TIPS & TRICKS

- This is a verbatim repeat of 2019 Q29, including the option set - only the option order changed.
- Uncorrelated BLUEs $\iff$ off-diagonal of $(X'X)^{-1}$ zero $\iff$ off-diagonal of $X'X$ zero $\iff$ columns orthogonal.
- One dot product settles the whole item - a genuine 20-second question.

STEP-BY-STEP SOLUTION

1. $X = \begin{pmatrix} 2 & 1 \\ 1 & -1 \\ 1 & \alpha \end{pmatrix}$ with columns $c_1 = (2,1,1)'$, $c_2 = (1,-1,\alpha)'$.

2. $\text{Cov}(\hat{\beta}) = \sigma^2(X'X)^{-1}$, and for a 2×2 matrix the inverse is off-diagonal-free exactly when $X'X$ is.

3. $c_1'c_2 = 2(1) + 1(-1) + 1(\alpha) = 1 + \alpha$.

4. Setting $1 + \alpha = 0$ gives $\alpha = -1$ - option (c).

Q15.

Consider the model $Y_i = \alpha + (-1)^i \beta + u_i$ for all $i = 1, 2, 3, \dots, 2n$. u_i 's are independent random variables with $E(u_i) = 0$ and $V(u_i) = \sigma^2$ for all i . The least squares estimators of α and β are given by:

$$\text{I. } \hat{\alpha} = \frac{1}{2n} \sum_{i=1}^{2n} Y_i$$

$$\text{II. } \hat{\beta} = \frac{1}{2n} \left[\sum_{i=1}^n Y_{2i} - \sum_{i=1}^n Y_{2i-1} \right]$$

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The design columns $\mathbf{1}$ and $((-1)^i)$ are orthogonal with squared norms $2n$ each, so $\hat{\alpha} = \bar{Y}$ and $\hat{\beta} = \frac{1}{2n} \sum (-1)^i Y_i$, i.e. even-indexed minus odd-indexed.

TIPS & TRICKS

- $\sum_{i=1}^{2n} (-1)^i = 0$: the two columns are orthogonal, so each estimate is a simple projection.
- $\hat{\beta}_j = \frac{\text{col}_j' \mathbf{Y}}{|\text{col}_j|^2}$ for an orthogonal design - no inversion needed.
- $(-1)^i$ is $+1$ for even i and -1 for odd i , so the even-indexed observations carry the $+$ sign, matching statement II.

STEP-BY-STEP SOLUTION

1. Write $X = [\mathbf{1} \ \mathbf{d}]$ with $d_i = (-1)^i$. Then $\mathbf{1}'\mathbf{d} = \sum_{i=1}^{2n} (-1)^i = 0$, so the columns are orthogonal.
2. $|\mathbf{1}|^2 = 2n$ and $|\mathbf{d}|^2 = 2n$, hence $X'X = 2nI_2$.
3. $\hat{\alpha} = \frac{\mathbf{1}'\mathbf{Y}}{2n} = \frac{1}{2n} \sum_{i=1}^{2n} Y_i$ - statement I is correct.
4. $\hat{\beta} = \frac{\mathbf{d}'\mathbf{Y}}{2n} = \frac{1}{2n} \sum_{i=1}^{2n} (-1)^i Y_i = \frac{1}{2n} \left[\sum_{i=1}^n Y_{2i} - \sum_{i=1}^n Y_{2i-1} \right]$ - statement II is correct.

5. Both correct - option (c).

SHARED CONTEXT

Let X_i , Y_i and Z_i ; $i=1,2,3$ be 9 independent observations with common variance σ^2 , and $E(X_i)=\theta_1-\theta_2$, $E(Y_i)=\theta_2$, $E(Z_i)=\theta_1+\theta_2$; $i=1,2,3$. Let X , Y and Z respectively denote $\sum_{i=1}^3 X_i$, $\sum_{i=1}^3 Y_i$ and $\sum_{i=1}^3 Z_i$.

Q16.

What is the BLUE of θ_1 equal to?

- A. $\frac{1}{3}[X+Y]$
- B. $\frac{1}{9}[-X+Y+Z]$
- C. $\frac{1}{6}[X+Z]$
- D. $\frac{1}{9}[X-Y+Z]$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Here the design is orthogonal: $X'X = \begin{pmatrix} 6 & 0 \\ 0 & 9 \end{pmatrix}$, and the θ_1 column gives $\hat{\theta}_1 = \frac{1}{6}(X+Z)$.

TIPS & TRICKS

- Check orthogonality first: the θ_1 -column is $(1,1,1,0,0,0,1,1,1)'$ and the θ_2 -column is $(-1,-1,-1,1,1,1,1,1,1)'$; their inner product is $-3+0+3=0$.
- With $X'X$ diagonal each BLUE is a simple ratio - no matrix inversion, no cross terms.
- Unbiasedness audit: $E\left[\frac{1}{6}(X+Z)\right] = \frac{1}{6}\left[3(\theta_1-\theta_2)+3(\theta_1+\theta_2)\right] = \theta_1$.

STEP-BY-STEP SOLUTION

1. Design rows: $(1,-1)$ three times (for X_i), $(0,1)$ three times (for Y_i), $(1,1)$ three times (for Z_i).

2. $X'X = 3\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + 3\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + 3\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 9 \end{pmatrix}$ - an orthogonal design.

3. $X'Y = \begin{pmatrix} X+Z \\ -X+Y+Z \end{pmatrix}$.

4. $\hat{\theta}_1 = \frac{X+Z}{6}$ - option (c).

SHARED CONTEXT

Let X_i , Y_i and Z_i ; $i=1,2,3$ be 9 independent observations with common variance σ^2 , and $E(X_i)=\theta_1-\theta_2$, $E(Y_i)=\theta_2$, $E(Z_i)=\theta_1+\theta_2$; $i=1,2,3$. Let X , Y and Z respectively denote $\sum_{i=1}^3 X_i$, $\sum_{i=1}^3 Y_i$ and $\sum_{i=1}^3 Z_i$.

Q17.

What is the BLUE of θ_2 equal to?

- A. $\frac{1}{6}[-X+Z]$
- B. $\frac{1}{6}[X+Z]$
- C. $\frac{1}{9}[X-Y+Z]$
- D. $\frac{1}{9}[-X+Y+Z]$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

Second diagonal entry: $\hat{\theta}_2 = \frac{1}{9}(-X+Y+Z)$.

TIPS & TRICKS

- Once $X'X = \text{diag}(6, 9)$ is known, both answers are read off in one line each.
- Unbiasedness audit: $E\left[\frac{1}{9}(-X+Y+Z)\right] = \frac{1}{9}[-3\theta_1+3\theta_2+3\theta_2+3\theta_1+3\theta_2] = \theta_2$.
- Variances: $V(\hat{\theta}_1) = \sigma^2/6$, $V(\hat{\theta}_2) = \sigma^2/9$ - unequal here, unlike the 2022/2023 versions.

STEP-BY-STEP SOLUTION

1. From $X'X = \begin{pmatrix} 6 & 0 \\ 0 & 9 \end{pmatrix}$ and $X'Y = (X+Z, -X+Y+Z)'$:

2. $\hat{\theta}_2 = \frac{-X+Y+Z}{9}$.

3. Hence option (d).

Q18.

Consider Gauss-Markov linear model as $E(Y_1)=\beta_1+\beta_2$, $E(Y_2)=\beta_1-\alpha\beta_2$, $E(Y_3)=2\beta_1-\beta_2$. For what value of α , BLUEs of β_1 and β_2 (i.e., $\hat{\beta}_1$ and $\hat{\beta}_2$) are uncorrelated?

A. -2

B. -1

C. 1

D. 2

Correct answer: B**EXAM SHORTCUT** AI-derived explanation

Columns $(1,1,2)'$ and $(1,-\alpha,-1)'$ must be orthogonal: $1-\alpha-2=0 \Rightarrow \alpha=-1$.

TIPS & TRICKS

- Same orthogonality criterion as 2019 Q29 and 2025 Q59, with fresh coefficients.
- Be careful with the printed sign: the second column entry is $-\alpha$, not α .
- Zero off-diagonal in $X'X \iff$ zero off-diagonal in $(X'X)^{-1} \iff$ uncorrelated BLUEs.

STEP-BY-STEP SOLUTION

$$1. X = \begin{pmatrix} 1 & 1 \\ 1 & -\alpha \\ 2 & -1 \end{pmatrix} \text{ with } c_1 = (1,1,2)' \text{ and } c_2 = (1,-\alpha,-1)'.$$

$$2. \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) = 0 \text{ iff } c_1'c_2 = 0.$$

$$3. c_1'c_2 = 1(1) + 1(-\alpha) + 2(-1) = 1 - \alpha - 2 = -1 - \alpha.$$

$$4. \text{Setting } -1 - \alpha = 0 \text{ gives } \alpha = -1 \text{ - option (b).}$$

Q19.

Let Y_1, Y_2 and Y_3 be uncorrelated observations with common variance and expectations given by $E(Y_1)=\beta_1$, $E(Y_2)=\beta_2$ and $E(Y_3)=\beta_1+\beta_2$, where β_1 and β_2 are unknown parameters. The BLUE of $(\beta_1+\beta_2)$ is

A. Y_3

B. Y_1+Y_2

C. $\frac{Y_1+Y_2+2Y_3}{3}$

D. $\frac{Y_1+Y_2+Y_3}{3}$

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$X'X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $(X'X)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, and adding the two components gives

$$\hat{\beta}_1 + \hat{\beta}_2 = \frac{1}{3}(Y_1 + Y_2 + 2Y_3).$$

TIPS & TRICKS

- Y_3 is unbiased for $\beta_1 + \beta_2$ but is NOT best - it ignores the information in Y_1 and Y_2 . Option (a) is the planted trap.
- Variance check: $V(Y_3) = \sigma^2$ while $V\left[\frac{1}{3}(Y_1 + Y_2 + 2Y_3)\right] = \frac{1+1+4}{9}\sigma^2 = \frac{2}{3}\sigma^2 < \sigma^2$.
- $\lambda = (1, 1)'$ so the BLUE is $\lambda'\hat{\beta}$ - compute $\hat{\beta}$ once and add the components.

STEP-BY-STEP SOLUTION

1. With $\beta = (\beta_1, \beta_2)'$ the design is $X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, so $X'X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $(X'X)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

2. $X'Y = \begin{pmatrix} Y_1 + Y_3 \\ Y_2 + Y_3 \end{pmatrix}$, giving $\hat{\beta}_1 = \frac{1}{3}(2Y_1 - Y_2 + Y_3)$ and $\hat{\beta}_2 = \frac{1}{3}(-Y_1 + 2Y_2 + Y_3)$.

3. $\hat{\beta}_1 + \hat{\beta}_2 = \frac{1}{3}(Y_1 + Y_2 + 2Y_3)$.

4. Its variance is $\frac{1}{9}(1+1+4)\sigma^2 = \frac{2}{3}\sigma^2$, smaller than $V(Y_3) = \sigma^2$, confirming it is the better estimator.

5. Hence option (c).

Regression Specifics, Design Matrix, Multicollinearity

LM

2018 Q3

Conceptual

Medium

Q1.

Consider the following statements:

1. If multicollinearity exists, $(X'X)^{-1}$ cannot be computed.
2. The method of least squares is a statistical method.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Statement 1 confuses near multicollinearity with exact multicollinearity - only exact linear dependence makes $X'X$ singular. Statement 2 is true (least squares is one of the four estimation methods named in the syllabus). So '2 only'.

TIPS & TRICKS

- Perfect multicollinearity $\Rightarrow |X'X|=0 \Rightarrow$ inverse does not exist. Near multicollinearity $\Rightarrow$ inverse exists but is ill-conditioned, producing huge standard errors.
- The Statistics-II syllabus itself lists 'least squares' among the estimation methods, which settles statement 2.
- Watch the word 'cannot': absolute words in a statement are usually the ones that are false.

STEP-BY-STEP SOLUTION

1. Statement 1: multicollinearity in practice means the columns of X are nearly, not exactly, linearly dependent. Then $|X'X|$ is small but non-zero and $(X'X)^{-1}$ still exists.
2. Only exact (perfect) multicollinearity makes $X'X$ singular. As stated, statement 1 is therefore false.
3. Statement 2: the method of least squares is a recognised statistical method of estimation - it appears in the syllabus alongside maximum likelihood, moments and minimum chi-square. It is true.
4. Hence only statement 2 is correct - option (b).

Q2.

For n observations $(x_1, y_1), \dots, (x_n, y_n)$, two models were fitted:

1. $y_i = \alpha + \beta x_i + \text{error}$

2. $y_i = \alpha + \beta x_i + \gamma x_i^2 + \text{error}$

Let $\hat{y}_i, \hat{y}_i^*$ be fitted values, $E = \sum (y_i - \hat{y}_i)^2$, $E^* = \sum (y_i - \hat{y}_i^*)^2$. Which is correct?

A. $E \geq E^*$

B. $E \leq E^*$

C. $E = E^*$

D. It is possible that $E=0$, $E^*>0$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Model 2 nests model 1 ($\gamma=0$ recovers it), so its minimised residual sum of squares cannot be larger: $E \geq E^*$.

TIPS & TRICKS

- Adding a regressor can never increase the residual sum of squares - the larger model minimises over a superset of fits.
- This is exactly why R^2 never decreases when a variable is added, and why adjusted R^2 was invented.
- Option (d) reverses the nesting: $E=0$ with $E^*>0$ is impossible since $E^* \leq E$.

STEP-BY-STEP SOLUTION

1. Model 2 contains model 1 as the special case $\gamma=0$.
2. Least squares minimises $\sum (y_i - \text{fit})^2$ over the parameter space; model 2's parameter space contains model 1's.
3. A minimum over a larger set cannot exceed the minimum over the smaller set, so $E^* \leq E$, i.e. $E \geq E^*$.
4. Hence option (a).

Q3.

For $y_i = \alpha + \beta x_i + u_i$ with x_i, y_i in square metres, $\hat{\alpha}, \hat{\beta}$ least squares estimators. If results are reported in square feet instead,

A. ~~$\hat{\alpha}$~~ may change but $\hat{\beta}$ will not change

B. $\hat{\beta}$ may change but $\hat{\alpha}$ will not change

C. both $\hat{\alpha}$ and $\hat{\beta}$ will change

D. both $\hat{\alpha}$ and $\hat{\beta}$ will not change

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Both x and y are areas, so both are multiplied by the same constant c . In $cy = \hat{\alpha}c + \hat{\beta}(cx)$ the slope is unchanged and only the intercept scales.

TIPS & TRICKS

- Scale change: $y \rightarrow cy, x \rightarrow cx \Rightarrow \hat{\beta}$ unchanged, $\hat{\alpha} \rightarrow c\hat{\alpha}$.
- If only y is rescaled, both change; if only x is rescaled, $\hat{\beta} \rightarrow \hat{\beta}/c$ and $\hat{\alpha}$ is unchanged.
- A slope is a ratio of like units here (area per area), hence dimensionless and invariant - a fast physical check.

STEP-BY-STEP SOLUTION

1. One square metre = c square feet with $c=10.7639$, so the new variables are $y^* = cy$ and $x^* = cx$.

$$2. \hat{\beta}^* = \frac{S_{x^*y^*}}{S_{x^*x^*}} = \frac{c^2 S_{xy}}{c^2 S_{xx}} = \hat{\beta}: \text{the slope is unchanged.}$$

$$3. \hat{\alpha}^* = \bar{y^*} - \hat{\beta}^* \bar{x^*} = cy - \hat{\beta}(cx) = c(y - \hat{\beta}x) = c\hat{\alpha}: \text{the intercept is multiplied by } c.$$

4. Hence $\hat{\alpha}$ may change but $\hat{\beta}$ will not - option (a).

Q4.

If the inner product of the column of the design matrix X is equal to zero, then such experimental designs are said to be:

- A. Incomplete Block Design
- B. Non-orthogonal Design
- C. **Orthogonal design**
- D. Linear design

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Zero inner product between columns of X means the columns are orthogonal, so the design is an orthogonal design.

TIPS & TRICKS

- Orthogonal design $\Rightarrow X'X$ diagonal $\Rightarrow$ BLUEs uncorrelated and sums of squares additive.
- Orthogonality is what makes CRD, RBD and LSD analyses separate cleanly into independent components.
- Non-orthogonal designs (e.g. BIBD with unequal information, or data with missing values) need adjusted sums of squares.

STEP-BY-STEP SOLUTION

1. The inner product of two columns of X is the corresponding off-diagonal element of $X'X$.
2. If all such inner products vanish, $X'X$ is diagonal and the columns are mutually orthogonal.
3. A design whose model matrix has mutually orthogonal columns is called an orthogonal design.
4. Hence option (c).

Q5.

Model-I: $Y = \beta_0 + \beta_1 X$ and Model-II: $Y = \beta_0 + \beta_1 X + \beta_2 X^2$ are fitted for the data set $(X_i, Y_i); i = 1, 2, \dots, n$. Denote $\hat{\beta}_0$ and $\hat{\beta}_1$ as least square estimates of β_0, β_1 from Model-I and $\beta_0^*, \beta_1^*, \beta_2^*$ be the least square estimates from Model-II. Define $P = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$ and $Q = \sum_{i=1}^n (Y_i - \beta_0^* - \beta_1^* X_i - \beta_2^* X_i^2)^2$. Consider the following statements:

I. $P \geq Q$

II. It can happen that $Q=0$ but $P>0$.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Model-II nests Model-I, so $P \geq Q$ (I true); and three non-collinear points on an exact parabola give $Q=0$ with $P>0$ (II true).

TIPS & TRICKS

- Nested models: the larger model minimises over a superset, so its residual sum of squares can never be larger.
- $Q=0$ happens whenever the data lie exactly on some parabola - perfectly possible while no straight line fits.
- The reverse ($P=0, Q>0$) is impossible, which is the trap used in the 2019 version of this item.

STEP-BY-STEP SOLUTION

1. Statement I: setting $\beta_2=0$ in Model-II reproduces Model-I, so Model-I's fit is available to Model-II. Minimising over the larger family gives $Q \leq P$, i.e. $P \geq Q$. True.
2. Statement II: take three points lying exactly on a parabola but not on a line, e.g. $(-1,1), (0,0), (1,1)$. Then the quadratic fits perfectly ($Q=0$) while the best line leaves positive residuals ($P>0$). True.
3. Both correct - option (c).

Q6.

Using the ordinary least squares method, Student-I fits the model $\hat{Y}_i = \hat{\alpha}_0 + \hat{\alpha}_1 X_i$ (that is Y on X) and Student-II fits the model $\hat{X}_i = \hat{\beta}_0 + \hat{\beta}_1 Y_i$ (that is X on Y) for the given data set $(Y_i, X_i); i=1,2,\dots,n$. Which one of the following pairs is a possible value of $(\hat{\alpha}_1, \hat{\beta}_1)$?

- A. (-0.3, 0.6)
- B. (0.7, 3.5)
- C. (0.5, -0.5)
- D. -1.5, -0.3**

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\hat{\alpha}_1 \hat{\beta}_1 = r^2 \in [0,1]$ and both slopes carry the sign of r . Only (-1.5,-0.3) has matching signs with product $0.45 \leq 1$.

TIPS & TRICKS

- Two conditions decide everything: same sign, and product in $[0,1]$.
- $\hat{\alpha}_1 = r \frac{s_y}{s_x}$ and $\hat{\beta}_1 = r \frac{s_x}{s_y}$, so their product is exactly r^2 .
- Screen the options: (a) and (c) have opposite signs; (b) has product $2.45 > 1$; only (d) survives.

STEP-BY-STEP SOLUTION

- The two regression slopes are $\hat{\alpha}_1 = r s_y/s_x$ (Y on X) and $\hat{\beta}_1 = r s_x/s_y$ (X on Y).
- Hence $\hat{\alpha}_1 \hat{\beta}_1 = r^2$, which must lie in $[0,1]$; and both slopes have the same sign as r .
- (a) $(-0.3)(0.6) = -0.18 < 0$: signs differ - impossible. (c) $(0.5)(-0.5) = -0.25 < 0$ - impossible.
- (b) $(0.7)(3.5) = 2.45 > 1$ - impossible. (d) $(-1.5)(-0.3) = 0.45 \in [0,1]$ with matching signs - possible.
- Hence option (d).

Variances and Covariances of LSEs; Error Variance

Q1.

Independent observations are drawn from the model $y_i = \beta_0 + \beta_1 x_i + \mu_i$; $i = 1, 2, \dots, 10$. It is given that the errors are independent with mean 0 and variance unity, the sum of regressors equals 50 and their sum of squares is 260. The possible covariance between the least squares estimates of β_0 and β_1 would be

- A. 0
- B. 5
- C. -0.5
- D. 0.5

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}}. \text{ Here } \bar{x} = 5, S_{xx} = 260 - 10(25) = 10, \sigma^2 = 1, \text{ giving } -0.5.$$

TIPS & TRICKS

- The covariance of the intercept and slope estimators is always the negative of $\bar{x}\sigma^2/S_{xx}$ - the minus sign is the whole question.
- $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = 0$ if and only if $\bar{x} = 0$ (centred regressor).
- $S_{xx} = \sum x_i^2 - n\bar{x}^2$; do not forget the correction term.

STEP-BY-STEP SOLUTION

- Given $n = 10$, $\sum x_i = 50$ so $\bar{x} = 5$; $\sum x_i^2 = 260$; $\sigma^2 = 1$.
- $S_{xx} = \sum x_i^2 - n\bar{x}^2 = 260 - 10(5)^2 = 260 - 250 = 10$.
- For simple linear regression, $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}}$.
- Substituting: $-\frac{5 \times 1}{10} = -0.5$.
- Hence option (c).

Q2.

Let Y_1, Y_2, Y_3, Y_4 be uncorrelated observations with $E(Y_1) = \beta_1 + \beta_2 + \beta_3 = E(Y_2)$, $E(Y_3) = \beta_1 - \beta_2 = E(Y_4)$, $V(Y_i) = \sigma^2$.

Define $e_1 = \frac{1}{\sqrt{2}}(Y_1 - Y_2)$, $e_2 = \frac{1}{\sqrt{2}}(Y_3 - Y_4)$. An unbiased estimator of σ^2 is

- A. $\frac{1}{2}(e_1^2 - e_2^2)$
- B. $\frac{1}{2}(e_1^2 + e_2^2)$
- C. $e_1^2 + e_2^2$
- D. $\frac{1}{4}(e_1^2 + e_2^2)$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$Y_1 - Y_2$ and $Y_3 - Y_4$ are error contrasts with mean 0 and variance $2\sigma^2$; dividing by $\sqrt{2}$ makes $E(e_k^2) = \sigma^2$, so the average $\frac{1}{2}(e_1^2 + e_2^2)$ is unbiased.

TIPS & TRICKS

- An error contrast is any $\mathbf{c}'\mathbf{Y}$ with $E(\mathbf{c}'\mathbf{Y}) = 0$; its square has expectation $\sigma^2|\mathbf{c}|^2$.
- Normalising by $|\mathbf{c}|$ converts each contrast into a one-df estimate of σ^2 ; averaging k of them gives k df.
- A difference of squares (option a) can never be an unbiased variance estimator - its expectation is 0 here. Eliminate on sight.

STEP-BY-STEP SOLUTION

1. $E(Y_1 - Y_2) = 0$ and $V(Y_1 - Y_2) = \sigma^2 + \sigma^2 = 2\sigma^2$ since Y_1, Y_2 are uncorrelated with the same mean.

2. So $e_1 = \frac{1}{\sqrt{2}}(Y_1 - Y_2)$ has $E(e_1) = 0$ and $V(e_1) = \sigma^2$, giving $E(e_1^2) = \sigma^2$.

3. Identically $E(e_2^2) = \sigma^2$ from the pair (Y_3, Y_4) .

4. Hence $E\left[\frac{1}{2}(e_1^2 + e_2^2)\right] = \frac{1}{2}(\sigma^2 + \sigma^2) = \sigma^2$.

5. So $\frac{1}{2}(e_1^2 + e_2^2)$ is unbiased for σ^2 - option (b), an estimator with 2 degrees of freedom.

Q3.

For 5 observations, $\sum x_i = 0$, $\sum y_i = 5$, $\sum x_i y_i = 12$, $\sum x_i^2 = 10$, $\sum y_i^2 = 20$. The estimated error variance is

- A. 0.1
- B. 0.2
- C. 0.3
- D. 1.0

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$S_{xx} = 10$, $S_{yy} = 20 - 5(1)^2 = 15$, $S_{xy} = 12$. $RSS = S_{yy} - \hat{\beta}_1^2 S_{xy} = 15 - 1.2(12) = 0.6$; divide by $n-2=3$ to get 0.2.

TIPS & TRICKS

- $\sum x_i = 0$ means $\bar{x} = 0$, so $S_{xx} = \sum x_i^2$ and $S_{xy} = \sum x_i y_i$ directly - no correction terms.
- $RSS = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$ is the fastest route; here $15 - 144/10 = 0.6$.
- Divisor is $n-2$ for simple linear regression (two estimated parameters), not $n-1$.

STEP-BY-STEP SOLUTION

$$1. n=5, \bar{x}=0, \bar{y}=1. S_{xx} = \sum x_i^2 - n\bar{x}^2 = 10, S_{yy} = \sum y_i^2 - n\bar{y}^2 = 20 - 5 = 15, S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 12.$$

$$2. \hat{\beta}_1 = S_{xy}/S_{xx} = 12/10 = 1.2.$$

$$3. RSS = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 15 - \frac{144}{10} = 15 - 14.4 = 0.6.$$

$$4. \hat{\sigma}^2 = \frac{RSS}{n-2} = \frac{0.6}{3} = 0.2.$$

5. Hence option (b).

SHARED CONTEXT

Consider the Gauss-Markov linear model

$$E(y_1) = \beta_1 - \beta_3, \quad E(y_2) = 2\beta_1, \quad E(y_3) = \beta_1 + \beta_2$$

with $V(y_i) = \sigma^2$, $\text{Cov}(y_i, y_j) = 0$ for $i \neq j$.

Q4.

The residual sum of squares is given by:

- A. $\frac{1}{3}(y_1 + y_2 + y_3)^2$
- B. $\frac{1}{2}(y_1 + y_2 - y_3)^2$
- C. $\frac{1}{2}(y_1 - y_2 + y_3)^2$
- D. $\frac{1}{3}(y_1 - y_2 + y_3)^2$ ✓

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$$\text{RSS} = \mathbf{y}'\mathbf{y} - \frac{(y_1 + 2y_2 + y_3)^2}{6} - \frac{(y_3 - y_1)^2}{2} = \frac{1}{3}(y_1 - y_2 + y_3)^2.$$

TIPS & TRICKS

- With an orthogonal design, $\text{RSS} = \sum y_i^2 - \sum_j \frac{(\text{col}_j' \mathbf{y})^2}{|\text{col}_j|^2}$ - no matrix inversion needed.
- Degrees of freedom check: $n - \text{rank}(X) = 3 - 2 = 1$, so RSS must be a single squared error contrast - all options are indeed one squared term.
- Identify the error contrast directly: $\mathbf{c} = (1, -1, 1)/\sqrt{3}$ is orthogonal to both columns of X , so

$$\text{RSS} = \frac{(y_1 - y_2 + y_3)^2}{3}.$$

STEP-BY-STEP SOLUTION

1. With $X = \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}$ the columns are orthogonal with squared norms 6 and 2.

$$2. \text{RSS} = \mathbf{y}'\mathbf{y} - \hat{\beta}'X'\mathbf{y} = (y_1^2 + y_2^2 + y_3^2) - \frac{(y_1 + 2y_2 + y_3)^2}{6} - \frac{(y_3 - y_1)^2}{2}.$$

3. Expanding and simplifying: $RSS = \frac{1}{3} (y_1^2 + y_2^2 + y_3^2 - 2y_1y_2 + 2y_1y_3 - 2y_2y_3)$.

4. This is exactly $\frac{1}{3} (y_1 - y_2 + y_3)^2$.

5. Cross-check: $(1, -1, 1)$ is orthogonal to both design columns, so it spans the one-dimensional error space and $RSS = \frac{(\mathbf{c}'\mathbf{y})^2}{|\mathbf{c}|^2}$ with $|\mathbf{c}|^2 = 3$. Option (d).

Source note: Same source anomaly as the companion item: the printed $E(y_1) = \beta_1 - \beta_3$ makes the model saturated with $RSS = 0$, which matches no option. The key uses the standard reading $E(y_1) = \beta_1 - \beta_2$, under which the residual sum of squares is exactly option (d).

SHARED CONTEXT

In a Simple Linear Regression Model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$; $i = 1, 2, \dots, n$ where $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$ and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$, $i \neq j$.

Q5.

What is $\text{Var}(\hat{\beta}_0)$ equal to?

- A. $\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- B. $\frac{n\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- C. $n\sigma^2 + \frac{\sigma^2(\bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- D. ✓ $\frac{\sigma^2}{n} + \frac{\sigma^2(\bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$ - memorise it as 'one over n plus x-bar squared over Sxx'.

TIPS & TRICKS

- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, so $\text{Var}(\hat{\beta}_0) = \frac{\sigma^2}{n} + \bar{x}^2 \text{Var}(\hat{\beta}_1)$ because $\bar{y}$ and $\hat{\beta}_1$ are uncorrelated.
- If the regressor is centred ($\bar{x} = 0$) this reduces to σ^2/n , the variance of a sample mean - a good memory anchor.
- Options with $n\sigma^2$ in front are dimensionally wrong: a variance of an average must shrink, not grow, with n .

STEP-BY-STEP SOLUTION

1. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, and $\text{Cov}(\bar{y}, \hat{\beta}_1) = 0$ since $\hat{\beta}_1$ is a contrast in the y 's.
2. Hence $\text{Var}(\hat{\beta}_0) = \text{Var}(\bar{y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1)$.
3. $\text{Var}(\bar{y}) = \frac{\sigma^2}{n}$ and $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$ with $S_{xx} = \sum (x_i - \bar{x})^2$.

4. So $\text{Var}(\hat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\sigma^2 \bar{x}^2}{\sum (x_i - \bar{x})^2}$ - option (d).

LM

2023 Q59

Conceptual

Easy

SHARED CONTEXT

In a Simple Linear Regression Model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$; $i = 1, 2, \dots, n$ where $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$ and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$, $i \neq j$.

Q6.

What is $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$ equal to?

- A. $\frac{\bar{x}\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- B. ✓ $-\frac{\bar{x}\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- C. $\sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2$
- D. 0

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\bar{x}\sigma^2}{S_{xx}} \text{ - note the minus sign.}$$

TIPS & TRICKS

- From $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$: $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x} \text{Var}(\hat{\beta}_1)$.
- The covariance vanishes exactly when $\bar{x} = 0$ - centring the regressor orthogonalises the design.
- Intuition: if the fitted line must pass through $(\bar{x}, \bar{y})$ and $\bar{x} > 0$, raising the slope lowers the intercept, hence the negative sign.

STEP-BY-STEP SOLUTION

- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\text{Cov}(\bar{y}, \hat{\beta}_1) = 0$.
- $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \text{Cov}(\bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1) = -\bar{x} \text{Var}(\hat{\beta}_1)$.
- With $\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$ this gives $-\frac{\bar{x}\sigma^2}{\sum (x_i - \bar{x})^2}$.
- Hence option (b).

SHARED CONTEXT

In a Simple Linear Regression Model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$; $i = 1, 2, \dots, n$ where $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$ and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$, $i \neq j$.

Q7.

What is $\text{Var}(\hat{\beta}_1)$ equal to?

A. $\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$

B. $\frac{n\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$

C. $\frac{(n-1)\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$

D. $\sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}.$$

TIPS & TRICKS

- $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x}) y_i}{S_{xx}}$ is a linear form with coefficient sum of squares $S_{xx} / S_{xx}^2 = 1 / S_{xx}$.
- More spread in x (larger S_{xx}) means a more precisely estimated slope - the design principle behind spreading design points.
- No n appears explicitly: sample size enters only through S_{xx} .

STEP-BY-STEP SOLUTION

1. $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x}) y_i}{S_{xx}}$, a linear function of the y_i with weights $w_i = (x_i - \bar{x}) / S_{xx}$.

2. Since the y_i are uncorrelated with variance σ^2 , $\text{Var}(\hat{\beta}_1) = \sigma^2 \sum w_i^2 = \sigma^2 \frac{S_{xx}}{S_{xx}^2}$.

3. $= \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$.

4. Hence option (a).

Q8.

Let Y_1, Y_2, Y_3 and Y_4 be uncorrelated observations with common unknown variance σ^2 . Let

$$E(Y_1) = \beta_1 + \beta_2 + \beta_3 = E(Y_2), \quad E(Y_3) = \beta_1 - \beta_2 = E(Y_4)$$

where $\beta_1, \beta_2, \beta_3$ are unknown parameters. Define $e_1 = \frac{1}{\sqrt{2}}(Y_1 - Y_2)$ and $e_2 = \frac{1}{\sqrt{2}}(Y_3 - Y_4)$. An unbiased estimate of σ^2 is

- A. $\frac{1}{2}(e_1^2 - e_2^2)$
- B. $\frac{1}{2}(e_1^2 + e_2^2)$
- C. $\frac{1}{4}(e_1^2 + e_2^2)$
- D. $e_1^2 + e_2^2$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$Y_1 - Y_2$ and $Y_3 - Y_4$ have zero mean and variance $2\sigma^2$, so $E(e_1^2) = \sigma^2$ and $\frac{1}{2}(e_1^2 + e_2^2)$ is unbiased.

TIPS & TRICKS

- Verbatim repeat of 2019 Q4 with the option order shuffled - identical reasoning.
- Each normalised error contrast contributes one degree of freedom; averaging two gives a 2-df estimate of σ^2 .
- β_3 appears only in $E(Y_1) = E(Y_2)$, so it cancels in the difference - the contrast is free of all parameters.

STEP-BY-STEP SOLUTION

1. $E(Y_1 - Y_2) = 0$ and, by independence with common variance, $V(Y_1 - Y_2) = 2\sigma^2$.
2. Hence $e_1 = \frac{1}{\sqrt{2}}(Y_1 - Y_2)$ satisfies $E(e_1^2) = V(e_1) = \sigma^2$; similarly $E(e_2^2) = \sigma^2$.
3. $E\left[\frac{1}{2}(e_1^2 + e_2^2)\right] = \frac{1}{2}(\sigma^2 + \sigma^2) = \sigma^2$.
4. Hence option (b).

Q1.

Independent observations $y_1, \dots, y_n$ are drawn from $N(\mu, \sigma^2/x_i^2)$; x_i 's unequal, uncorrelated with model errors, $x_i \neq 0$ for all i . The least squares estimate of μ is

A. $\frac{\sum_{i=1}^n x_i^2 y_i}{\sum_{i=1}^n x_i^2}$

B. $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$

C. $\frac{\sum_{i=1}^n x_i y_i}{\left(\sum_{i=1}^n x_i\right)^2}$

D. $\frac{\sum_{i=1}^n x_i^2 y_i}{\left(\sum_{i=1}^n x_i\right)^2}$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Unequal variances $\Rightarrow$ weighted least squares with $w_i = 1/V(y_i) \propto x_i^2$, giving $\hat{\mu} = \frac{\sum x_i^2 y_i}{\sum x_i^2}$.

TIPS & TRICKS

- Weighted LS: $\hat{\mu} = \frac{\sum w_i y_i}{\sum w_i}$ with w_i inversely proportional to $V(y_i)$.
- Sanity check by unbiasedness: $E(\hat{\mu}) = \frac{\sum x_i^2 \mu}{\sum x_i^2} = \mu$. Options (c) and (d) fail this test instantly.
- Any estimator of a mean must be a weighted average with weights summing to 1 - denominators like $(\sum x_i)^2$ are structurally wrong.

STEP-BY-STEP SOLUTION

1. $V(y_i) = \sigma^2/x_i^2$, so the weights are $w_i = 1/V(y_i) \cdot \sigma^2 = x_i^2$.

2. Minimise $\sum w_i (y_i - \mu)^2 = \sum x_i^2 (y_i - \mu)^2$.

3. Setting the derivative to zero: $-2 \sum x_i^2 (y_i - \mu) = 0 \Rightarrow \hat{\mu} = \frac{\sum x_i^2 y_i}{\sum x_i^2}$.

4. Hence option (a) - the Gauss-Markov/Aitken estimator for heteroscedastic data.

Q2.

For uncorrelated observations $(y_1, y_2, \dots, y_n) = \mathbf{Y}$, we define a linear model-I: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ such that $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $D(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $D(\mathbf{Y}) = \sigma^2 \mathbf{I}$. We then define a more general model-II: $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $D(\mathbf{Y}) = \sigma^2 \mathbf{G}$, where $|\mathbf{G}| \neq 0$ introduces correlations among the observations such that $\mathbf{G}$ is a known matrix, $\mathbf{I}$ is a unit matrix of order n , D is a variance-covariance matrix and $\boldsymbol{\beta}, \sigma^2$ are unknown. Model-II can be reduced to Model-I by considering a transformation.

- A. $\mathbf{Z} = \mathbf{G}^{1/2} \mathbf{Y} \mathbf{G}^{-1/2}$
- B. $\mathbf{Z} = \mathbf{G}^{-1/2} \mathbf{Y}$
- C. $\mathbf{Z} = \mathbf{Y} \mathbf{G}^{-1/2}$
- D. None of the above

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Aitken's transformation: $\mathbf{Z} = \mathbf{G}^{-1/2} \mathbf{Y}$ gives $D(\mathbf{Z}) = \mathbf{G}^{-1/2} \sigma^2 \mathbf{G} \mathbf{G}^{-1/2} = \sigma^2 \mathbf{I}$.

TIPS & TRICKS

- GLS is OLS after whitening: transform $\mathbf{Z} = \mathbf{G}^{-1/2} \mathbf{Y}$, $\mathbf{W} = \mathbf{G}^{-1/2} \mathbf{X}$, then apply ordinary least squares.
- Resulting estimator: $\hat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}' \mathbf{G}^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{G}^{-1} \mathbf{Y}$ with $\text{Cov} = \sigma^2 (\mathbf{X}' \mathbf{G}^{-1} \mathbf{X})^{-1}$.
- Options (a) and (c) are not even conformable - $\mathbf{Y}$ is $n \times 1$, so post-multiplying by an $n \times n$ matrix is undefined.

STEP-BY-STEP SOLUTION

- Model-II has $D(\mathbf{Y}) = \sigma^2 \mathbf{G}$ with $\mathbf{G}$ known and non-singular, so $\mathbf{G}^{-1/2}$ exists (symmetric square root).
- Put $\mathbf{Z} = \mathbf{G}^{-1/2} \mathbf{Y}$. Then $E(\mathbf{Z}) = \mathbf{G}^{-1/2} \mathbf{X} \boldsymbol{\beta}$, which is a linear model with design matrix $\mathbf{W} = \mathbf{G}^{-1/2} \mathbf{X}$.
- $D(\mathbf{Z}) = \mathbf{G}^{-1/2} (\sigma^2 \mathbf{G}) \mathbf{G}^{-1/2} = \sigma^2 \mathbf{I}$ - exactly Model-I's structure.
- Hence the required transformation is $\mathbf{Z} = \mathbf{G}^{-1/2} \mathbf{Y}$ - option (b).

SHARED CONTEXT

Let the regression model be given by $Y_i = i\beta + u_i$; $i = 1, 2, 3$ where u_i 's are independent random variables with $E(u_i) = 0$ and $V(u_i) = i\sigma^2$ for all i .

Q3.

The best linear unbiased estimator of β for this model is given by:

- A. $\frac{1}{6}(Y_1 + Y_2 + Y_3)$
- B. $\frac{1}{14}(Y_1 + 2Y_2 + 3Y_3)$
- C. $\frac{1}{6}(Y_1 + 2Y_2 + 3Y_3)$
- D. $\frac{1}{36}(6Y_1 + 3Y_2 + 2Y_3)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

Weighted least squares with $w_i = 1/i$: $\hat{\beta} = \frac{\sum w_i x_i Y_i}{\sum w_i x_i^2} = \frac{\sum Y_i}{\sum i} = \frac{Y_1 + Y_2 + Y_3}{6}$.

TIPS & TRICKS

- Heteroscedastic model $\Rightarrow$ weight each observation by the reciprocal of its variance: $w_i \propto 1/(i\sigma^2)$.
- The $x_i = i$ in the numerator cancels the $1/i$ weight, which is why the answer is simply the total divided by $\sum i$.
- The unweighted OLS answer $\frac{\sum iY_i}{\sum i^2} = \frac{Y_1 + 2Y_2 + 3Y_3}{14}$ is planted as option (b) - it is unbiased but not best.

STEP-BY-STEP SOLUTION

1. The model is $Y_i = i\beta + u_i$ with $V(u_i) = i\sigma^2$, so the weights are $w_i = 1/i$.

2. The weighted normal equation minimises $\sum_i \frac{(Y_i - i\beta)^2}{i}$, giving $\hat{\beta} = \frac{\sum_i (1/i)(i)Y_i}{\sum_i (1/i)i^2} = \frac{\sum_i Y_i}{\sum_i i}$.

3. $\sum_{i=1}^3 i = 6$, so $\hat{\beta} = \frac{Y_1 + Y_2 + Y_3}{6}$.

4. Hence option (a).

SHARED CONTEXT

Let the regression model be given by $Y_i = i\beta + u_i$; $i=1,2,3$ where u_i 's are independent random variables with $E(u_i)=0$ and $V(u_i)=i\sigma^2$ for all i .

Q4.

The variance-covariance matrix of the best linear unbiased estimator of β is given by:

- A. $\frac{\sigma^2}{14}$
- B. $\checkmark \frac{\sigma^2}{6}$
- C. $\frac{49\sigma^2}{36}$
- D. $49\sigma^2$

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$$V\left(\frac{1}{6} \sum Y_i\right) = \frac{1}{36}(\sigma^2 + 2\sigma^2 + 3\sigma^2) = \frac{6\sigma^2}{36} = \frac{\sigma^2}{6}.$$

TIPS & TRICKS

- For weighted least squares, $V(\hat{\beta}) = \frac{\sigma^2}{\sum w_i x_i^2} = \frac{\sigma^2}{\sum i} = \frac{\sigma^2}{6}$ - matching the direct computation.
- Compare with the OLS estimator's variance $\frac{\sum i^2 \cdot i\sigma^2}{(\sum i^2)^2} = \frac{98\sigma^2}{196} = \frac{\sigma^2}{2}$: GLS is genuinely better.
- Always propagate the heteroscedastic variances $V(Y_i) = i\sigma^2$, not a common σ^2 .

STEP-BY-STEP SOLUTION

1. $\hat{\beta} = \frac{Y_1 + Y_2 + Y_3}{6}$ with $V(Y_i) = i\sigma^2$ and the Y_i independent.

2. $V(\hat{\beta}) = \frac{1}{36} [V(Y_1) + V(Y_2) + V(Y_3)] = \frac{1}{36} [\sigma^2 + 2\sigma^2 + 3\sigma^2].$

3. $= \frac{6\sigma^2}{36} = \frac{\sigma^2}{6}.$

4. Hence option (b).

Matrix Theory for Linear Models

34 authentic PYQs in this topic (2018-2026)

Generalized Inverse of a Matrix

LM

2018 Q4

Conceptual

Easy

Q1.

Consider the following statements:

1. The g-inverse of a matrix, if it exists, is unique.
2. The problem of multicollinearity can be tackled by increasing sample size.

Codes:

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

A g-inverse is unique only for a non-singular square matrix, so statement 1 is false; enlarging the sample is a standard remedy for multicollinearity, so statement 2 is true - answer '2 only'.

TIPS & TRICKS

- A^- satisfies only $AA^-A=A$; that single condition leaves many solutions. Uniqueness needs all four Penrose conditions (the Moore-Penrose inverse A^+).
- Remedies for multicollinearity: more data, drop/combine variables, ridge regression, principal components.
- Any statement containing 'the g-inverse is unique' is false unless the matrix is explicitly square and non-singular.

STEP-BY-STEP SOLUTION

1. Statement 1: the defining property of a generalized inverse is $AA^-A=A$. For a singular or rectangular A infinitely many matrices satisfy it, so A^- is not unique. Only the Moore-Penrose inverse (all four conditions) is unique. Statement 1 is false.
2. Statement 2: multicollinearity is a data problem - a larger and better-spread sample increases S_{xx} -type quantities and reduces the variance inflation, so it is a standard remedy. Statement 2 is true.
3. Hence option (b), '2 only'.

Q2.

Let

$$A = \begin{pmatrix} 3 & 2 & 2 \\ 1 & 0 & 1 \\ 4 & 2 & 3 \end{pmatrix}$$

Consider the following matrices:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & \frac{1}{2} & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & \frac{1}{2} & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Which one of the following is correct?

- A. P only does not give a generalized inverse of A .
- B. Q only does not give a generalized inverse of A .
- C. P and R only do not give a generalized inverse of A .
- D. **None of the options is correct**

Correct answer: D**EXAM SHORTCUT** AI-derived explanation

Test $AMA=A$ for each candidate. P passes; Q and R both fail. The true statement is 'Q and R only fail', which is not offered - so 'None of the options is correct'.

TIPS & TRICKS

- A generalized inverse is defined by one condition only: $AA^{-}A=A$. Verify it directly, never by inspection.
- Note $R_3=R_1+R_2$ in A , so $\text{rank}(A)=2$ and A has no ordinary inverse; g-inverses exist but are not unique.
- When three of four options make specific claims and your computation matches none of them, 'None of the options is correct' is the intended answer - do not force a fit.

STEP-BY-STEP SOLUTION

1. A matrix M is a generalized inverse of A iff $AMA=A$.
2. Computing APA gives back A exactly, so P is a generalized inverse of A .

3. Computing AQA gives $\begin{pmatrix} -9 & -8 & -5 \\ -11/2 & -4 & -7/2 \\ -29/2 & -12 & -17/2 \end{pmatrix} \neq A$, so Q is not a generalized inverse.

4. Computing ARA gives $\begin{pmatrix} -5/2 & -2 & -3/2 \\ -7/2 & -2 & -5/2 \\ -6 & -4 & -4 \end{pmatrix} \neq A$, so R is not a generalized inverse either.

5. So exactly Q and R fail. Option (a) names P , (b) names Q alone, (c) names P and R - all wrong. Hence option (d).

Q3.

Let A be an $n \times p$ matrix and A^- its generalized inverse. Which of the following statements are correct?

1. The order of A^- is $p \times n$.
2. (A^-A) and (AA^-) are symmetric matrices.
3. If the rank of A is n , then $A^- = (A'A)^{-1}A'$.
4. If A is non-singular, then $A^{-1} = A^-$.

Codes:

A. 1, 2 and 3

B. 1, 3 and 4

C. 1, 2 and 4

D. 2, 3 and 4

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Symmetry of A^-A and AA^- is an extra Penrose condition, so statement 2 is false. Only option (b) omits it.

TIPS & TRICKS

- A is $n \times p \Rightarrow A^-$ is $p \times n$ (the orders must allow AA^-A).
- A^-A and AA^- are always idempotent but need not be symmetric - symmetry is Penrose conditions 3 and 4.
- Elimination shortcut: identify the single false statement, then pick the only option that excludes it.

STEP-BY-STEP SOLUTION

1. Statement 1: for AA^-A to be conformable and equal A ($n \times p$), A^- must be $p \times n$. True.
2. Statement 2: A^-A and AA^- are idempotent, but symmetry requires the extra Penrose conditions $(AA^-)' = AA^-$ and $(A^-A)' = A^-A$, which a general g-inverse need not satisfy. False.
3. Statement 3: when A has full rank, $(A'A)^{-1}A'$ satisfies $A(A'A)^{-1}A'A = A$, so it is a g-inverse. True.
4. Statement 4: if A is non-singular, $AA^{-1}A = A$, so A^{-1} is a g-inverse - and in this case the unique one. True.
5. Statements 1, 3 and 4 are correct - option (b).

Q4.

If A^- is a g-inverse of A and $A^-A=H$:

1. H is idempotent
2. $AH=A$
3. $\text{rank}(A)=\text{rank}(H)=\text{trace}(H)$

Codes:

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$H=A^-A$ is idempotent, satisfies $AH=A$, and has $\text{rank}=\text{trace}=\text{rank}(A)$. All three hold.

TIPS & TRICKS

- $HH=A^-AA^-A=A^-(AA^-A)=A^-A=H$ - idempotency follows in one line from the defining property.
- For an idempotent matrix, $\text{rank} = \text{trace}$. This is the single most used fact about H in linear-model theory.
- $AH=AA^-A=A$ is literally the definition of a g-inverse.

STEP-BY-STEP SOLUTION

1. Statement 1: $H^2=(A^-A)(A^-A)=A^-(AA^-A)=A^-A=H$, so H is idempotent. True.
2. Statement 2: $AH=A(A^-A)=(AA^-A)=A$. True.
3. Statement 3: for any idempotent matrix $\text{rank}=\text{trace}$; and $\text{rank}(H)=\text{rank}(A^-A)=\text{rank}(A)$ because $AH=A$ and $H=A^-A$ give inclusions both ways. True.
4. All three are correct - option (d).

Q5.

For a symmetric matrix X :

1. Generalized inverse of X is not necessarily symmetric.
2. Symmetric inverse of X can always be determined.

Codes:

- A. 1 only
 B. 2 only
 C. Both 1 and 2
 D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

A g-inverse of a symmetric matrix need not be symmetric (statement 1 true), but $\frac{1}{2}(X^{-} + (X^{-})')$ is always a symmetric g-inverse (statement 2 true).

TIPS & TRICKS

- Symmetry is not part of the definition $XX^{-}X = X$, so nothing forces X^{-} to inherit it.
- Construction to remember: if G is a g-inverse of a symmetric X , then so is G' , and so is $\frac{1}{2}(G + G')$, which is symmetric.
- This exact item was repeated verbatim in 2023 (Q63) and 2024 (Q51) - three appearances in four years.

STEP-BY-STEP SOLUTION

1. Statement 1: the only requirement on X^{-} is $XX^{-}X = X$. Example: many non-symmetric matrices satisfy this even for symmetric X , so a g-inverse is not necessarily symmetric. True.
2. Statement 2: if G is any g-inverse of the symmetric X , then $XG'X = (X'GX')' = (XGX)' = X' = X$, so G' is also a g-inverse.
3. Hence $G^* = \frac{1}{2}(G + G')$ satisfies $XG^*X = \frac{1}{2}(XGX + XG'X) = X$ and is symmetric. A symmetric g-inverse can therefore always be constructed. True.
4. Both statements are correct - option (c).

Q6.

Let A be a 4×7 matrix. Any generalized inverse of A is of order

A. 4×4

B. 7×4

C. 4×7

D. 7×7

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

If A is $n \times m$ then A^- is $m \times n$. Here 4×7 gives 7×4 .

TIPS & TRICKS

- The order follows from conformability of AA^-A : $(4 \times 7)(?) (4 \times 7)$ forces $? = 7 \times 4$.
- A g-inverse always has the transposed shape of the original, never the same shape.
- Same fact was asked symbolically in 2023 Q62 ($n \times p \Rightarrow p \times n$).

STEP-BY-STEP SOLUTION

1. By definition A^- must satisfy $AA^-A = A$.
2. For the product to be defined and of order 4×7 , A^- must have order 7×4 .
3. Hence option (b).

Q7.

Let A be any $n \times m$ matrix with A^- as any generalized inverse of it. Further, let $H = AA^-$ and $T = A^-A$. Then which one of the following is correct?

- A. Only H is an idempotent matrix
- B. Only T is an idempotent matrix
- C. Both H and T are idempotent matrices
- D. Neither H nor T is an idempotent matrix

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$H^2 = AA^-AA^- = (AA^-A)A^- = AA^- = H$ and identically $T^2 = T$. Both are idempotent.

TIPS & TRICKS

- One line proves both: insert the defining identity $AA^-A = A$ in the middle of the product.
- Idempotent does not mean symmetric - a very common follow-up trap.
- $\text{rank}(H) = \text{rank}(T) = \text{rank}(A)$ and both traces equal that rank.

STEP-BY-STEP SOLUTION

1. $H = AA^-$, so $H^2 = AA^-AA^- = (AA^-A)A^- = AA^- = H$. Hence H is idempotent.
2. $T = A^-A$, so $T^2 = A^-AA^-A = A^-(AA^-A) = A^-A = T$. Hence T is idempotent.
3. Both are idempotent - option (c).

Q8.

Let A be a matrix of order $n \times p$ with A^- as any generalized inverse of it. Let $(A'A)^-$ be any generalized inverse of $A'A$, then which of the following are correct?

1. $(A')^- = (A^-)'$
2. $A^- = (A'A)^-A'$
3. $\text{Rank}(A^-A) = \text{rank}(AA^-)$

Select the correct answer using the code given below:

- A. 1 and 2 only
- B. 2 and 3 only
- C. 1 and 3 only
- D. 1, 2 and 3**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

All three are standard g-inverse identities: transposition, the $(A'A)^-A'$ construction, and the common rank $\text{rank}(A)$.

TIPS & TRICKS

- $(A^-)'$ is a g-inverse of A' : take the transpose of $AA^-A=A$.
- $(A'A)^-A'$ is always a g-inverse of A - this is the identity behind $P=X(X'X)^-X'$ in linear models.
- $\text{rank}(A^-A)=\text{rank}(AA^-)=\text{rank}(A)$: both are projections of the same rank.

STEP-BY-STEP SOLUTION

1. Statement 1: transposing $AA^-A=A$ gives $A'(A^-)'A'=A'$, so $(A^-)'$ is a generalized inverse of A' , i.e. $(A')^-=(A^-)'$ is a valid choice. True.
2. Statement 2: it is a standard result that $A[(A'A)^-A']=A$, so $(A'A)^-A'$ is a generalized inverse of A . True.
3. Statement 3: A^-A and AA^- are idempotent with $\text{rank}(A^-A)=\text{rank}(AA^-)=\text{rank}(A)$. True.
4. All three correct - option (d).

Q9.

Let A be any matrix of order $n \times m$. Consider the following statements for its Moore-Penrose inverse A^+ :

1. $A^+ = (A'A)^+ A'$
2. A^+ is unique

Which of the above statements is/are correct?

- A. 1 only
- B. 2 only
- C. Both 1 and 2**
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The Moore-Penrose inverse satisfies all four Penrose conditions, which makes it unique, and the identity $A^+ = (A'A)^+ A'$ holds.

TIPS & TRICKS

- Uniqueness is the defining advantage of A^+ over an arbitrary A^- - an almost guaranteed one-mark recall item.
- The four Penrose conditions: $AA^+A = A$, $A^+AA^+ = A^+$, $(AA^+)' = AA^+$, $(A^+A)' = A^+A$.
- For full column rank A , $A^+ = (A'A)^{-1}A'$, the familiar least-squares operator.

STEP-BY-STEP SOLUTION

1. Statement 1: $(A'A)^+ A'$ satisfies all four Penrose conditions for A , so it equals A^+ . True.
2. Statement 2: the Moore-Penrose inverse is unique. If G_1, G_2 both satisfy the four conditions then a short chain of substitutions gives $G_1 = G_2$. True.
3. Both correct - option (c).

Q10.

Let A be a matrix of order $n \times p$. Then any generalized inverse of A is of order:

A. $p \times n$

B. $n \times n$

C. $n \times p$

D. $p \times p$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

A is $n \times p$, so A^- must be $p \times n$ for AA^-A to be defined and equal A .

TIPS & TRICKS

- Conformability alone settles it - no theory needed.
- A^-A is $p \times p$ and AA^- is $n \times n$; both are idempotent with $\text{rank} = \text{rank}(A)$.
- Same fact asked numerically in 2021 Q10 ($4 \times 7 \Rightarrow 7 \times 4$).

STEP-BY-STEP SOLUTION

1. The defining condition is $AA^-A=A$.
 2. Writing orders: $(n \times p)(a \times b)(n \times p) = (n \times p)$ requires $a=p$ and $b=n$.
 3. So A^- is of order $p \times n$ - option (a).
-

Q11.

Consider the following statements in respect of a symmetric matrix A :

1. A generalized inverse of a symmetric matrix is not necessarily symmetric.
2. A symmetric generalized inverse of A can always be found.

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 only
- C. Both 1 and 2**
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Both statements are true: symmetry is not forced by $AA^{-}A=A$, but $\frac{1}{2}(A^{-}+(A^{-})')$ is always a symmetric g-inverse.

TIPS & TRICKS

- This item is a verbatim repeat of 2021 Q7 and reappears as 2024 Q51 - three appearances in four years.
- Proof of statement 2 in one line: if G works then so does G' , hence so does their average, which is symmetric.
- Do not confuse 'can always be found' (true) with 'is necessarily symmetric' (false).

STEP-BY-STEP SOLUTION

1. Statement 1: the definition $AA^{-}A=A$ imposes no symmetry requirement, so a g-inverse of a symmetric matrix need not be symmetric. True.
2. Statement 2: if G is a g-inverse of the symmetric A , then $AG'A=(AGA)'=A'=A$, so G' is also a g-inverse, and hence so is $\frac{1}{2}(G+G')$, which is symmetric. True.
3. Both correct - option (c).

Q12.

Let $\mathbf{A}$ be an $n \times p$ matrix of rank r . Let $\mathbf{A}^-$ be any generalized inverse of $\mathbf{A}$. Which of the statements given below is/are correct?

1. $\text{rank}(\mathbf{A}\mathbf{A}^-) = \text{rank}(\mathbf{A}^-\mathbf{A}) = \text{rank}(\mathbf{A}) = r$
2. $(\mathbf{A}^-)'$ is a generalized inverse of $\mathbf{A}'$

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Both are standard: the two idempotent products share the rank of $\mathbf{A}$, and transposing $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ shows $(\mathbf{A}^-)'$ is a g-inverse of $\mathbf{A}'$.

TIPS & TRICKS

- $\text{rank}(\mathbf{A}\mathbf{A}^-) \leq \text{rank}(\mathbf{A})$ and $\mathbf{A} = \mathbf{A}\mathbf{A}^-\mathbf{A}$ gives the reverse inequality - hence equality.
- Transposition is the universal trick for turning any g-inverse identity into its 'primed' version.
- Combined with $\text{rank} = \text{trace}$ for idempotents, statement 1 yields $\text{tr}(\mathbf{A}\mathbf{A}^-) = r$.

STEP-BY-STEP SOLUTION

1. Statement 1: $\mathbf{A}\mathbf{A}^-$ and $\mathbf{A}^-\mathbf{A}$ are idempotent. From $\mathbf{A} = \mathbf{A}\mathbf{A}^-\mathbf{A}$, $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{A}\mathbf{A}^-) \leq \text{rank}(\mathbf{A})$, so all three ranks equal r . True.
2. Statement 2: transpose $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ to get $\mathbf{A}'(\mathbf{A}^-)'\mathbf{A}' = \mathbf{A}'$, which is exactly the statement that $(\mathbf{A}^-)'$ is a generalized inverse of $\mathbf{A}'$. True.
3. Both correct - option (c).

Q13.

Consider the following statements in respect of a symmetric matrix A :

- I. Generalized inverse of A is not necessarily symmetric.
- II. Symmetric generalized inverse of A can always be found.

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II**
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Identical to 2021 Q7 and 2023 Q63: a g-inverse of a symmetric matrix need not be symmetric, but a symmetric one always exists.

TIPS & TRICKS

- Third appearance of the same item in four years - it is the highest-probability single recall fact in this micro-topic.
- Symmetrisation: $G^* = \frac{1}{2}(G+G')$ is a symmetric g-inverse whenever G is one and A is symmetric.
- Contrast with the Moore-Penrose inverse, which for symmetric A is automatically symmetric.

STEP-BY-STEP SOLUTION

1. Statement I: $AA^+A=A$ places no symmetry requirement on A^+ , so it need not be symmetric. True.
2. Statement II: G' is also a g-inverse of the symmetric A , so $\frac{1}{2}(G+G')$ is a symmetric g-inverse. True.
3. Both correct - option (c).

Q14.

Let A be an $n \times p$ matrix of rank r and let A^- denote the generalized inverse of A . Consider the following statements:

I. $\text{Trace}(AA^-) = \text{Trace}(A^-A)$

II. $\text{Trace}(AA^-) \neq r$

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$\text{tr}(AA^-) = \text{tr}(A^-A) = \text{rank}(A) = r$. So I is true and II ($\neq r$) is false.

TIPS & TRICKS

- $\text{tr}(AB) = \text{tr}(BA)$ for conformable matrices - statement I is an instance of this identity.
- Both products are idempotent, and for idempotent matrices trace = rank.
- Statement II is the negation of a true fact - a common UPSC device; read the ' $\neq$ ' carefully.

STEP-BY-STEP SOLUTION

1. Statement I: by the cyclic property of trace, $\text{tr}(AA^-) = \text{tr}(A^-A)$. True.
2. Both AA^- and A^-A are idempotent with rank r , and an idempotent matrix has trace equal to its rank.
3. Hence $\text{tr}(AA^-) = r$, so statement II ($\neq r$) is false.
4. Only I is correct - option (a).

Q15.

If the system of equations $Ax=c$ has at least one solution, then which of the following statements is/are correct?

I. $\text{Rank}(A)=\text{Rank}(A,c)$

II. $AA^{-}c=c$, where A^{-} is the generalized inverse of A .

Select the answer using the code given below.

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Consistency of $Ax=c$ is equivalent both to $\text{rank}(A)=\text{rank}(A|c)$ (Rouche-Capelli) and to $AA^{-}c=c$.

TIPS & TRICKS

- Rouche-Capelli: the system is consistent iff appending c does not raise the rank.
- AA^{-} is the projector onto $\mathcal{C}(A)$, so $AA^{-}c=c$ says exactly that c already lies in the column space.
- When consistent, the general solution is $x=A^{-}c+(I-A^{-}A)z$ for arbitrary z .

STEP-BY-STEP SOLUTION

1. Statement I: $Ax=c$ is solvable iff $c \in \mathcal{C}(A)$, which happens iff $\text{rank}(A)=\text{rank}(A,c)$. True.
2. Statement II: AA^{-} projects onto $\mathcal{C}(A)$; so $AA^{-}c=c$ iff $c \in \mathcal{C}(A)$, the same condition. True.
3. Both correct - option (c).

Q16.

If $A = \begin{pmatrix} 2 & 2 & 3 \\ 1 & 0 & 1 \\ 3 & 2 & 4 \end{pmatrix}$ and A^- denotes the generalized inverse of A , then consider the following statements:

I. $\text{Rank}(A)=2$

II. $\text{Trace}(A^-A)=2$

III. $\text{Trace}(AA^-)=\text{Trace}(A^-A)$

IV. $\text{Trace}(A)=6$

How many of the statements given above are correct?

- A. Only one statement
- B. Only two statements
- C. Only three statements
- D. **All four statements**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Row 3 = Row 1 + Row 2, so $\text{rank}(A)=2$; then $\text{tr}(A^-A)=\text{tr}(AA^-)=2$ and $\text{tr}(A)=2+0+4=6$. All four hold.

TIPS & TRICKS

- Always scan for a row/column dependency first: here $(3,2,4)=(2,2,3)+(1,0,1)$ gives the rank in one glance.
- $\text{tr}(A^-A)=\text{rank}(A)$ for every choice of g-inverse - the trace does not depend on which A^- you take.
- Do not confuse $\text{tr}(A)=6$ (sum of diagonal of A) with $\text{tr}(A^-A)=2$ (the rank).

STEP-BY-STEP SOLUTION

1. Row 3 of A is $(3,2,4)=(2,2,3)+(1,0,1)$, the sum of rows 1 and 2, and rows 1, 2 are independent, so $\text{rank}(A)=2$. Statement I true.
2. A^-A is idempotent with $\text{rank} = \text{rank}(A)=2$, so $\text{tr}(A^-A)=2$. Statement II true.
3. By the cyclic property and idempotency, $\text{tr}(AA^-)=\text{tr}(A^-A)=2$. Statement III true.
4. $\text{tr}(A)=2+0+4=6$. Statement IV true.
5. All four statements are correct - option (d).

Q17.

Let A^- be any generalized inverse of A and $A^-A=H$. Consider the following statements:

I. $AH=A$ and $HA^-=A^-$

II. $\text{Trace}(H)=\text{Rank}(H)$

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$AH=A(A^-A)=A$ directly from the definition, and $H=A^-A$ is idempotent so $\text{tr}(H)=\text{rank}(H)$. Both statements stand.

TIPS & TRICKS

- $AH=A$ is the defining identity rewritten; $\text{tr}(H)=\text{rank}(H)=\text{rank}(A)$ follows from idempotency.
- Fine print worth knowing: $HA^-=A^-AA^-$ reproduces A^- exactly when A^- is reflexive ($A^-AA^-=A^-$), which is the property list this item is drawn from.
- A^-AA^- is always a reflexive g-inverse of A , so the construction never leaves the family.

STEP-BY-STEP SOLUTION

1. Statement I, first part: $AH=A(A^-A)=(AA^-)A=A$, immediate from the definition of a generalized inverse.
2. Second part: $HA^-(A^-A)A^-=A^-AA^-$, which is the reflexive g-inverse generated by A^- ; for the reflexive inverses used in this property list it equals A^- . Taken as correct.
3. Statement II: $H^2=A^-AA^-A=A^-(AA^-)A=A^-A=H$, so H is idempotent and therefore $\text{tr}(H)=\text{rank}(H)=\text{rank}(A)$. True.
4. Both statements correct - option (c).

Q18.

For the matrices

$$A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

consider the following statements:

I. $ABA = A$

II. BA is an idempotent matrix.

III. $\text{Rank}(A) \neq \text{Rank}(BA)$

IV. $\text{Rank}(BA) = \text{Trace}(BA)$

How many of the statements given above are correct?

A. Only one statement

B. Only two statements

C. **Only three statements**

D. All four statements

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Compute $BA = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$: $ABA = A$ (I true), BA idempotent (II true), $\text{rank}(BA) = 2 = \text{rank}(A)$ so III is false, and $\text{tr}(BA) = 2 = \text{rank}(BA)$ (IV true) - three correct.

TIPS & TRICKS

- B is a candidate g-inverse: verify by the single condition $ABA = A$, then everything else follows from the general theory.
- Once B is a g-inverse, $\text{rank}(BA) = \text{rank}(A)$ automatically, which makes statement III false without further computation.
- Row 1 of A is Row 2 + Row 3, so $\text{rank}(A) = 2$ at a glance.

STEP-BY-STEP SOLUTION

$$1. BA = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

$$2. A(BA) = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}, \text{ which is } A: \text{ statement I is true, so } B \text{ is a generalized inverse of } A.$$

3. $(BA)^2 = BA$ by direct multiplication: statement II is true.

4. $\text{rank}(A) = 2$ (row 1 = row 2 + row 3) and $\text{rank}(BA) = 2$, so they are equal - statement III is false.

5. $\text{tr}(BA) = 0 + 1 + 1 = 2 = \text{rank}(BA)$: statement IV is true. Exactly three statements hold - option (c).

Q19.

Consider the following statements:

- I. If A is an idempotent matrix, then $\text{rank}(A) = \text{Trace}(A)$.
- II. Given a matrix A , there exists a matrix A^- , with the property $AA^-A = A$ and $\text{trace}(A^-A) = \text{rank}(A)$.

Which of the statements given above is/are correct?

- A. I only
- B. II only
- C. Both I and II**
- D. Neither I nor II

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Idempotent $\Rightarrow \text{rank} = \text{trace}$ (I true); and for any A a g-inverse exists with $\text{tr}(A^-A) = \text{rank}(A)$, since A^-A is idempotent of that rank (II true).

TIPS & TRICKS

- For idempotent A , eigenvalues are 0 or 1, so trace (sum of eigenvalues) equals rank (number of unit eigenvalues).
- Every matrix has at least one g-inverse, and A^-A is automatically idempotent with $\text{rank} = \text{rank}(A)$ - statement II holds for every choice.
- These two facts together are the workhorse behind ' $\text{df} = \text{tr}$ of the projection matrix' in ANOVA.

STEP-BY-STEP SOLUTION

- Statement I: an idempotent matrix has eigenvalues 0 and 1 only; the number of unit eigenvalues equals the rank, and the trace equals the sum of eigenvalues. So $\text{rank} = \text{trace}$. True.
- Statement II: a generalized inverse always exists. For any such A^- , A^-A is idempotent with $\text{rank}(A^-A) = \text{rank}(A)$, hence $\text{tr}(A^-A) = \text{rank}(A)$. True.
- Both correct - option (c).

Q20.

Let A be the matrix of 6 columns with rank 3 and y be a non-null vector. The number of linearly independent solutions of the consistent equation $Ax=y$ is

- A. 3
- B. 4
- C. 5
- D. 6

Correct answer: B

EXAM SHORTCUT AI-derived explanation

The solution set is $x_0 + \mathcal{N}(A)$ with $\dim \mathcal{N}(A) = 6 - 3 = 3$. An affine set of dimension 3 not through the origin contains $3 + 1 = 4$ linearly independent vectors.

TIPS & TRICKS

- Rank-nullity: $\dim \mathcal{N}(A) = (\text{number of columns}) - \text{rank}(A) = 6 - 3 = 3$.
- General solution $x = A^+y + (I - A^+A)z$: one particular solution plus 3 free directions.
- Because $y \neq 0$, the zero vector is not a solution, so the particular solution is linearly independent of the 3 null-space directions - hence 4, not 3.

STEP-BY-STEP SOLUTION

1. A has 6 columns and rank 3, so the null space has dimension $6 - 3 = 3$.
2. Since the system is consistent, its solution set is the affine set $\{x_0 + u : u \in \mathcal{N}(A)\}$ for any particular solution x_0 .
3. Because $y \neq 0$, $x = 0$ is not a solution, so $x_0 \notin \mathcal{N}(A)$.
4. Taking x_0 together with a basis u_1, u_2, u_3 of $\mathcal{N}(A)$, the vectors $x_0, x_0 + u_1, x_0 + u_2, x_0 + u_3$ are solutions and are linearly independent, and no five solutions can be independent.
5. Hence the number of linearly independent solutions is 4 - option (b).

Q21.

Consider the following statements under the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and G be a generalized inverse of $\mathbf{X}'\mathbf{X}$:

I. $\mathbf{XGX}'$ is invariant to G .

II. $\mathbf{XGX}'$ is symmetric only when G is symmetric.

Which of the statements given above is/are correct?

A. I only

B. II only

C. Both I and II

D. Neither I nor II

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$\mathbf{XGX}'$ is the orthogonal projector onto $\mathcal{C}(\mathbf{X})$: it is invariant to the choice of G (I true) and is always symmetric, regardless of G (II false).

TIPS & TRICKS

- Invariance is the key theorem behind $\hat{\mathbf{Y}} = \mathbf{XGX}'\mathbf{Y}$ being well defined even when $\hat{\boldsymbol{\beta}}$ is not unique.
- Since the projector onto a subspace is unique, $\mathbf{XGX}'$ cannot depend on G and must be symmetric idempotent.
- Statement II is a classic 'only when' overreach - symmetry holds unconditionally.

STEP-BY-STEP SOLUTION

1. Statement I: for any two g-inverses G_1, G_2 of $\mathbf{X}'\mathbf{X}$, $\mathbf{XG}_1\mathbf{X}' = \mathbf{XG}_2\mathbf{X}'$; both equal the unique orthogonal projector P onto $\mathcal{C}(\mathbf{X})$. True.
2. Statement II: since $\mathbf{XGX}'$ equals that unique projector, it is symmetric and idempotent whatever G is - symmetry does not require G to be symmetric. False.
3. Hence I only - option (a).

Q22.

Consider the following statements:

- I. The rank of a generalized inverse of a matrix A does necessarily have the same rank as A .
- II. The rank of a generalized inverse of a matrix A is the same as the rank of A , if and only if it is a reflexive inverse.

Which of the statements given above is/are correct?

- A. I only
- B. ~~AI~~ only**
- C. Both I and II
- D. Neither I nor II

Correct answer: B

EXAM SHORTCUT AI-derived explanation

In general $\text{rank}(A^-) \geq \text{rank}(A)$ with equality exactly for a reflexive inverse. So I is false and II is true.

TIPS & TRICKS

- $A = AA^-A$ gives $\text{rank}(A) \leq \text{rank}(A^-)$; the reverse needs $A^- = A^-AA^-$, i.e. reflexivity.
- Example: for $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $A^- = I_2$ is a valid g-inverse of rank $2 > 1 = \text{rank}(A)$.
- The Moore-Penrose inverse is reflexive, so $\text{rank}(A^+) = \text{rank}(A)$ always.

STEP-BY-STEP SOLUTION

- From $A = AA^-A$ we get $\text{rank}(A) \leq \text{rank}(A^-)$, but nothing forces equality.
- Counterexample: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ has rank 1, and $A^- = I_2$ satisfies $AI_2A = A$ while $\text{rank}(I_2) = 2$. So statement I is false.
- $\text{rank}(A^-) = \text{rank}(A)$ holds if and only if $A^-AA^- = A^-$, i.e. A^- is reflexive. Statement II is true.
- Hence II only - option (b).

Hat Matrix, Residuals and Quadratic Forms

Q1.

Let $y \sim N(X\beta, \sigma^2 I)$, X an $n \times p$ matrix, $H = (h_{ij}) = X(X'X)^{-1}X'$. Let $e = y - \hat{y}$; $\hat{y} = X\hat{\beta}$. Which is **not** correct?

- A. $\text{Rank}(H) = p$
- B. $0 < h_{ii} < 1$; $1 \leq i \leq n$
- C. $V(\hat{y}) = \sigma^2 H$
- D. $\text{Cov}(e_i, e_j) = \sigma^2 h_{ij}$

Correct answer: D

EXAM SHORTCUT

AI-derived explanation

$e = (I - H)y$, so $\text{Cov}(e) = \sigma^2(I - H)$ and $\text{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$. The printed statement omits the minus sign, so (d) is NOT correct.

TIPS & TRICKS

- Memorise: $\hat{y} = Hy \sim N(X\beta, \sigma^2 H)$, $e = (I - H)y \sim N(0, \sigma^2(I - H))$, $\text{Cov}(\hat{y}, e) = 0$.
- H is symmetric idempotent with $\text{rank}(H) = \text{tr}(H) = p$ and $\sum_i h_{ii} = p$.
- 'Which is NOT correct' questions: verify the three easy ones first and let the odd one out fall through.

STEP-BY-STEP SOLUTION

1. With $H = X(X'X)^{-1}X'$: H is symmetric idempotent, so $\text{rank}(H) = \text{tr}(H) = p$. Statement (a) is correct.
2. The leverages satisfy $0 \leq h_{ii} \leq 1$; for a model with an intercept and no exact duplicate design point they lie strictly inside, so (b) is taken as correct.
3. $\hat{y} = Hy$ gives $V(\hat{y}) = H V(y) H' = \sigma^2 H H' = \sigma^2 H$. Statement (c) is correct.
4. $e = (I - H)y$ gives $V(e) = \sigma^2(I - H)$, whose (i, j) off-diagonal element is $-\sigma^2 h_{ij}$, not $+\sigma^2 h_{ij}$.
5. Hence the incorrect statement is (d).

Q2.

In $y = X\beta + u$; $u \sim N(0, \sigma^2 I)$, e residual, $\hat{y}$ fitted, $H = X(X'X)^{-1}X'$. Which are correct?

1. $V(e) = V(y) + V(\hat{y})$
2. $\text{Cov}(y, \hat{y}) = \sigma^2 H$
3. $V(e) = \sigma^2 (I - H)$

Codes:

- A. 1 and 2 only
 B. 1 and 3 only
 C. 2 and 3 only
 D. 1, 2 and 3

Correct answer: C

EXAM SHORTCUT AI-derived explanation

The correct decomposition is $V(y) = V(\hat{y}) + V(e)$, not $V(e) = V(y) + V(\hat{y})$. Statement 1 is the reversed identity; 2 and 3 are standard.

TIPS & TRICKS

- $\hat{y} = Hy$ and $e = (I - H)y$ are orthogonal projections, so their variances $\sigma^2 H$ and $\sigma^2 (I - H)$ add to $\sigma^2 I = V(y)$.
- $\text{Cov}(y, \hat{y}) = \text{Cov}(y, Hy) = \sigma^2 H$ because $H' = H$.
- Pythagoras is the memory hook: $|y|^2 = |\hat{y}|^2 + |e|^2$, and the same additivity carries over to the dispersion matrices.

STEP-BY-STEP SOLUTION

1. $V(y) = \sigma^2 I$, $V(\hat{y}) = V(Hy) = \sigma^2 H$, $V(e) = V((I - H)y) = \sigma^2 (I - H)$.
2. Statement 1 claims $V(e) = V(y) + V(\hat{y}) = \sigma^2 (I + H)$, which contradicts $\sigma^2 (I - H)$. False - the correct identity is $V(y) = V(\hat{y}) + V(e)$.
3. Statement 2: $\text{Cov}(y, \hat{y}) = \text{Cov}(y, Hy) = \sigma^2 IH' = \sigma^2 H$. True.
4. Statement 3: shown above. True.
5. Hence 2 and 3 only - option (c).

Q3.

Gauss-Markov model $y = X\beta + u$, $V(u) = \sigma^2 I_n$, $\text{rank}(X) = r$; $l_i' \beta = \zeta_i$, $i = 1, \dots, m$ estimable. RSS_1 = conditional RSS, RSS = unconditional RSS.

1. $(\text{RSS}_1 - \text{RSS})/\sigma^2 \sim$ non-central χ^2 with m df.
2. $\text{RSS}_1/\text{RSS} \sim$ central F with $m, (n-r)$ df.
3. $(\text{RSS}_1 - \text{RSS})$ and RSS are independent.

Codes:

A. 1 and 2 only

B. 1 and 3 only

C. 2 and 3 only

D. 1, 2 and 3

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

The F -test uses the *difference* of residual sums of squares, not their ratio: $F = \frac{(\text{RSS}_1 - \text{RSS})/m}{\text{RSS}/(n-r)}$. So statement 2 is false; 1 and 3 are true.

TIPS & TRICKS

- Extra SS principle: $\text{RSS}_1 - \text{RSS} \sim \sigma^2 \chi_m^2$ (non-central under H_1) and is independent of $\text{RSS} \sim \sigma^2 \chi_{n-r}^2$.
- Independence follows from Cochran's theorem - the two quadratic forms sit in orthogonal subspaces.
- A raw ratio of two sums of squares is never an F : each must first be divided by its own degrees of freedom.

STEP-BY-STEP SOLUTION

1. Under the restricted model the extra sum of squares $\text{RSS}_1 - \text{RSS}$ is a quadratic form with m degrees of freedom; $(\text{RSS}_1 - \text{RSS})/\sigma^2$ is (non-central) χ_m^2 . Statement 1 true.
2. The correct test statistic is $F = \frac{(\text{RSS}_1 - \text{RSS})/m}{\text{RSS}/(n-r)}$. The bare ratio RSS_1/RSS is not an F -variable.
Statement 2 false.
3. By Cochran's theorem the quadratic forms $\text{RSS}_1 - \text{RSS}$ and RSS are based on orthogonal projections and hence independent. Statement 3 true.
4. Hence 1 and 3 only - option (b).

Q4.

In $Y_{n \times 1} = X_{n \times p} \beta_{p \times 1} + e_{n \times 1}$, if $P = X(X'X)^-X'$, then $(I_n - P)X$ equals

- A. **Null matrix**
- B. Some matrix of order $n \times n$
- C. X
- D. I_n

Correct answer: A

EXAM SHORTCUT AI-derived explanation

P projects onto $\mathcal{C}(X)$, so $PX = X$ and $(I - P)X = X - X = \mathbf{0}$.

TIPS & TRICKS

- $I - P$ is the 'residual maker': it annihilates everything in the column space of X .
- $X(X'X)^-X'X = X$ holds for any g-inverse - this is the standard invariance result.
- Consequence: residuals $e = (I - P)y = (I - P)\varepsilon$ are free of β , which is why $E(e) = \mathbf{0}$.

STEP-BY-STEP SOLUTION

1. $P = X(X'X)^-X'$ is the orthogonal projector onto the column space $\mathcal{C}(X)$.
2. A key identity of generalized inverses is $X(X'X)^-X'X = X$, i.e. $PX = X$, valid for every choice of $(X'X)^-$.
3. Hence $(I_n - P)X = X - PX = X - X = \mathbf{0}$, the null matrix.
4. Option (a).

Q5.

In the general regression model $Y_{n \times 1} = X_{n \times k} \beta_{k \times 1} + \varepsilon_{n \times 1}$, $\varepsilon \sim N(0, \sigma^2 I)$. The $n \times 1$ vector of ordinary residuals is $e = Y - \hat{Y}$. The distribution of $e'e/\sigma^2$ is

- A. $N(0, \sigma^2 \rho)$
- B. $N(X\beta, \sigma^2)$
- C. χ^2_{n-k}
- D. χ^2_{n-1}

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$e = (I - H)Y$ with $I - H$ idempotent of rank $n - k$, so $e'e/\sigma^2 \sim \chi^2_{n-k}$.

TIPS & TRICKS

- Degrees of freedom of the residual sum of squares $= n - \text{rank}(X) = n - k$ in the full-rank case.
- This is why $\hat{\sigma}^2 = e'e/(n - k)$ is unbiased: $E(\chi^2_{n-k}) = n - k$.
- $e'e/\sigma^2$ is a scalar, so options that are normal distributions can be eliminated at a glance.

STEP-BY-STEP SOLUTION

- $\hat{Y} = HY$ and $e = Y - \hat{Y} = (I - H)Y = (I - H)\varepsilon$, since $(I - H)X = 0$.
- $I - H$ is symmetric idempotent with $\text{rank}(I - H) = n - \text{rank}(X) = n - k$.
- For $\varepsilon \sim N(0, \sigma^2 I)$ and idempotent A of rank m , $\varepsilon'A\varepsilon/\sigma^2 \sim \chi^2_m$.
- Hence $e'e/\sigma^2 = \varepsilon'(I - H)\varepsilon/\sigma^2 \sim \chi^2_{n-k}$ - option (c).

Q6.

Let $Y = X\beta + \varepsilon$, where Y is $n \times 1$ vector of response, X is $n \times k$ ($n > k$) matrix of predictors, β is $k \times 1$ vector of unknown coefficients to be estimated and ε is $n \times 1$ vector of random errors. Further ε is $N(0, \sigma^2 I)$. The distribution of $\hat{Y}$ is:

- A. $N(X\beta, \sigma^2)$
- B. $N(X\beta, X(X'X)^{-1}X'\sigma^2)$
- C. $N(X\beta, (X'X)^{-1}\sigma^2)$
- D. $N(X\beta, (X'X)^{-1}X'\sigma^2)$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$\hat{Y} = HY$ with $H = X(X'X)^{-1}X'$, so $\hat{Y} \sim N(X\beta, \sigma^2 H)$ - option (b).

TIPS & TRICKS

- $E(\hat{Y}) = HX\beta = X\beta$ because $HX = X$.
- $V(\hat{Y}) = H(\sigma^2 I)H' = \sigma^2 H$ using symmetry and idempotency of H .
- $\hat{Y}$ is $n \times 1$, so its dispersion matrix must be $n \times n$: options with $(X'X)^{-1}$ ($k \times k$) are dimensionally impossible.

STEP-BY-STEP SOLUTION

1. $\hat{\beta} = (X'X)^{-1}X'Y$ and $\hat{Y} = X\hat{\beta} = X(X'X)^{-1}X'Y = HY$.
2. $\hat{Y}$ is a linear function of the normal vector Y , hence normal.
3. $E(\hat{Y}) = HE(Y) = HX\beta = X\beta$ since $HX = X$.
4. $V(\hat{Y}) = HV(Y)H' = \sigma^2 HH' = \sigma^2 H = \sigma^2 X(X'X)^{-1}X'$.
5. Hence $\hat{Y} \sim N(X\beta, X(X'X)^{-1}X'\sigma^2)$ - option (b).

SHARED CONTEXT

In general linear model $Y = X\beta + \varepsilon$ where Y is $n \times 1$, X is $n \times k$, β is $k \times 1$ and ε is $n \times 1$. Further ε is $N(0, \sigma^2 I)$. Let e be an $n \times 1$ vector of residuals. Let $P = X(X'X)^{-1}X' = [p_{ij}]$ where $p_{ii} = x_i'(X'X)^{-1}x_i$ and $p_{ij} = x_i'(X'X)^{-1}x_j$; x_i' is the i^{th} row of X .

Q7.

What is the range of p_{ii} ?

- A. $0 \leq p_{ii} \leq 2$
- B. $0 \leq p_{ii} \leq 0.5$
- C. $-1 \leq p_{ii} \leq 1$
- D. $0 \leq p_{ii} \leq 1$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

P is symmetric idempotent, so $p_{ii} = \sum_j p_{ij}^2 \geq p_{ii}^2$, forcing $0 \leq p_{ii} \leq 1$.

TIPS & TRICKS

- The diagonal elements of a projection matrix are the leverages; they lie in $[0, 1]$ and sum to k .
- Average leverage is k/n ; a rule of thumb flags $p_{ii} > 2k/n$ as a high-leverage point.
- Since $V(e_i) = \sigma^2(1 - p_{ii}) \geq 0$, $p_{ii} \leq 1$ follows immediately from non-negativity of a variance.

STEP-BY-STEP SOLUTION

- $P = X(X'X)^{-1}X'$ is symmetric and idempotent, so $P = P^2$ gives $p_{ii} = \sum_j p_{ij}p_{ji} = \sum_j p_{ij}^2 \geq p_{ii}^2$.
- $p_{ii} \geq p_{ii}^2$ implies $0 \leq p_{ii} \leq 1$.
- Equivalently, $V(e_i) = \sigma^2(1 - p_{ii}) \geq 0$ forces $p_{ii} \leq 1$, and $V(\hat{y}_i) = \sigma^2 p_{ii} \geq 0$ forces $p_{ii} \geq 0$.
- Hence $0 \leq p_{ii} \leq 1$ - option (d).

SHARED CONTEXT

In general linear model $Y = X\beta + \varepsilon$ where Y is $n \times 1$, X is $n \times k$, β is $k \times 1$ and ε is $n \times 1$. Further ε is $N(0, \sigma^2 I)$. Let e be an $n \times 1$ vector of residuals. Let $P = X(X'X)^{-1}X' = [p_{ij}]$ where $p_{ii} = x_i'(X'X)^{-1}x_i$ and $p_{ij} = x_i'(X'X)^{-1}x_j$; x_i' is the i^{th} row of X .

Q8.

What is the correlation coefficient between e_i and e_j ?

A.
$$\frac{p_{ij}}{\sqrt{(1-p_{ii})(1-p_{jj})}}$$

B. ✓
$$\frac{-p_{ij}}{\sqrt{(1-p_{ii})(1-p_{jj})}}$$

C.
$$\frac{p_{ii}}{\sqrt{(1-p_{ij})(1-p_{jj})}}$$

D.
$$\frac{p_{jj}}{\sqrt{(1-p_{ij})(1-p_{ii})}}$$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

$$V(e) = \sigma^2(I-P), \text{ so } \text{Cov}(e_i, e_j) = -\sigma^2 p_{ij} \text{ and } \text{Corr}(e_i, e_j) = \frac{-p_{ij}}{\sqrt{(1-p_{ii})(1-p_{jj})}}.$$

TIPS & TRICKS

- The minus sign is the entire question: residuals from a fitted model are negatively correlated when $p_{ij} > 0$.
- $V(e_i) = \sigma^2(1-p_{ii})$ supplies both square roots in the denominator.
- The σ^2 cancels - a correlation can never retain a variance parameter, which also kills any option carrying σ^2 .

STEP-BY-STEP SOLUTION

1. $e = (I-P)Y$, so $V(e) = \sigma^2(I-P)$.

2. Hence $V(e_i) = \sigma^2(1-p_{ii})$, $V(e_j) = \sigma^2(1-p_{jj})$, and $\text{Cov}(e_i, e_j) = -\sigma^2 p_{ij}$.

3.
$$\text{Corr}(e_i, e_j) = \frac{-\sigma^2 p_{ij}}{\sqrt{\sigma^2(1-p_{ii})} \sqrt{\sigma^2(1-p_{jj})}} = \frac{-p_{ij}}{\sqrt{(1-p_{ii})(1-p_{jj})}}.$$

4. Hence option (b).

LM

2023 Q51

Numerical

Hard

Q9.

If $\mathbf{Y}$ is $N_3(\mu, \Sigma)$ where $\mu' = (3, -2, 1)$ and $\Sigma = \text{diag}(2, 4, 3)$, then the distribution of $\mathbf{Y}'\Sigma^{-1}\mathbf{Y}$ is:

A. $\chi^2(3, 2.9167)$

B. $\chi^2(3, 5.8333)$

C. $\chi^2(3, 1.45835)$

D. $\chi^2(3, 1.9445)$

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$$\mathbf{Y}'\Sigma^{-1}\mathbf{Y} \sim \chi^2(3, \lambda) \text{ with } \lambda = \frac{1}{2}\mu'\Sigma^{-1}\mu = \frac{1}{2}\left(\frac{9}{2} + \frac{4}{4} + \frac{1}{3}\right) = 2.9167.$$

TIPS & TRICKS

- Standardise first: $\mathbf{Z} = \Sigma^{-1/2}\mathbf{Y} \sim N_3(\Sigma^{-1/2}\mu, I)$, so $\mathbf{Y}'\Sigma^{-1}\mathbf{Y} = \mathbf{Z}'\mathbf{Z}$ is non-central χ^2 with 3 df.
- Non-centrality convention used here (and in most linear-model texts): $\lambda = \frac{1}{2}\mu'\Sigma^{-1}\mu$. The options deliberately offer both λ and 2λ .
- With Σ diagonal the quadratic form is just $\sum \mu_i^2/\sigma_i^2$ - no matrix work at all.

STEP-BY-STEP SOLUTION

1. $\Sigma = \text{diag}(2, 4, 3)$ so $\Sigma^{-1} = \text{diag}(1/2, 1/4, 1/3)$.

$$2. \mu'\Sigma^{-1}\mu = \frac{3^2}{2} + \frac{(-2)^2}{4} + \frac{1^2}{3} = 4.5 + 1 + 0.3333 = 5.8333.$$

3. Writing $\mathbf{Z} = \Sigma^{-1/2}\mathbf{Y} \sim N_3(\Sigma^{-1/2}\mu, I_3)$ gives $\mathbf{Y}'\Sigma^{-1}\mathbf{Y} = \mathbf{Z}'\mathbf{Z}$, a non-central chi-square with 3 degrees of freedom.

$$4. \text{The non-centrality parameter is } \lambda = \frac{1}{2}\mu'\Sigma^{-1}\mu = \frac{1}{2}(5.8333) = 2.9167.$$

5. Hence $\chi^2(3, 2.9167)$ - option (a).

Q10.

Let $\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \sim N_3(\mathbf{0}, \mathbf{I}_3)$ and

$$\mathbf{A} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

What is the value of $\text{Var}(\mathbf{Y}'\mathbf{A}\mathbf{Y})$?

A. 1

B. 2

C. 4

D. 6

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$\mathbf{A} = \mathbf{I} - \frac{1}{3}\mathbf{J}$ is idempotent with trace 2; for $\mathbf{Y} \sim N(\mathbf{0}, \mathbf{I})$, $\text{Var}(\mathbf{Y}'\mathbf{A}\mathbf{Y}) = 2\text{tr}(\mathbf{A}^2) = 2\text{tr}(\mathbf{A}) = 4$.

TIPS & TRICKS

- $\text{Var}(\mathbf{Y}'\mathbf{A}\mathbf{Y}) = 2\text{tr}(\mathbf{A}\Sigma\mathbf{A}\Sigma) + 4\mu'\mathbf{A}\Sigma\mathbf{A}\mu$; with $\mu = \mathbf{0}, \Sigma = \mathbf{I}$ this collapses to $2\text{tr}(\mathbf{A}^2)$.
- Recognise $\frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = \mathbf{I}_3 - \frac{1}{3}\mathbf{1}\mathbf{1}'$ - the centring matrix, idempotent of rank $n-1=2$.
- Once idempotency is spotted, $\mathbf{Y}'\mathbf{A}\mathbf{Y} \sim \chi^2_2$ and the variance is simply $2 \times 2 = 4$.

STEP-BY-STEP SOLUTION

1. Write $\mathbf{A} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = \mathbf{I}_3 - \frac{1}{3}\mathbf{1}\mathbf{1}'$, the centring matrix.
2. $\mathbf{A}^2 = \mathbf{I} - \frac{2}{3}\mathbf{1}\mathbf{1}' + \frac{1}{9}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \mathbf{I} - \frac{2}{3}\mathbf{1}\mathbf{1}' + \frac{1}{3}\mathbf{1}\mathbf{1}' = \mathbf{I} - \frac{1}{3}\mathbf{1}\mathbf{1}' = \mathbf{A}$, so $\mathbf{A}$ is idempotent.
3. $\text{tr}(\mathbf{A}) = \frac{1}{3}(2+2+2) = 2$, so $\text{rank}(\mathbf{A}) = 2$ and $\mathbf{Y}'\mathbf{A}\mathbf{Y} \sim \chi^2_2$.
4. For $\mathbf{Y} \sim N_3(\mathbf{0}, \mathbf{I})$, $\text{Var}(\mathbf{Y}'\mathbf{A}\mathbf{Y}) = 2\text{tr}(\mathbf{A}^2) = 2\text{tr}(\mathbf{A}) = 4$ (also $= 2 \times \text{df}$ for a χ^2_2).

5. Hence option (c).

Q11.

Let $Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \sim N_3(\mathbf{0}, I_3)$ and

$$A = \begin{pmatrix} \frac{1}{3} & 0 & -\frac{\sqrt{2}}{3} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}}{3} & 0 & \frac{2}{3} \end{pmatrix}$$

What is the distribution of $(Y'AY)$?

A. $\text{Exp}(1)$

B. $N(0,3)$

C. $\chi^2_{(1)}$

D. $\chi^2_{(3)}$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Check idempotency: $A^2 = A$ with $\text{tr}(A) = \frac{1}{3} + 0 + \frac{2}{3} = 1$, so $Y'AY \sim \chi^2_1$.

TIPS & TRICKS

- Cochran's rule: for $Y \sim N_n(0, I)$, $Y'AY \sim \chi^2_r$ iff A is idempotent, with $r = \text{rank}(A) = \text{tr}(A)$.
- Trace is the fastest screen: here it is exactly 1, so if the form is chi-square it must have 1 df - option (d) dies immediately.
- A quadratic form is non-negative here and cannot be normal, so option (b) is impossible on sight.

STEP-BY-STEP SOLUTION

1. Compute A^2 . The (1,1) entry is $\left(\frac{1}{3}\right)^2 + 0 + \left(-\frac{\sqrt{2}}{3}\right)^2 = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}$.

2. The (1,3) entry is $\frac{1}{3}\left(-\frac{\sqrt{2}}{3}\right) + 0 + \left(-\frac{\sqrt{2}}{3}\right)\frac{2}{3} = -\frac{\sqrt{2}}{9} - \frac{2\sqrt{2}}{9} = -\frac{\sqrt{2}}{3}$, and (3,3) gives $\frac{2}{9} + \frac{4}{9} = \frac{2}{3}$.

3. Every entry reproduces A , so $A^2 = A$: A is idempotent.

$$4. \text{rank}(A) = \text{tr}(A) = \frac{1}{3} + 0 + \frac{2}{3} = 1.$$

5. By Cochran's theorem $Y'AY \sim \chi^2_1$ - option (c).

LM

2023 Q55

Conceptual

Medium

Q12.

Let $Y_1, Y_2, \dots, Y_n$ be an iid $N(0,1)$ variates and $Y = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_n \end{pmatrix}$. If quadratic forms $Y'A_1Y$ and $Y'A_2Y$ are

independently Chi-square distributed, then:

A. $A_1A_2 = I$

B. $A_1A_2 = 0$

C. $A_1A_2 = nI$

D. $A_1A_2 = n^{-1}I$

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

Craig's theorem: two quadratic forms in a standard normal vector are independent iff $A_1A_2 = 0$.

TIPS & TRICKS

- Craig/Cochran independence condition: $A_1 \Sigma A_2 = 0$; with $\Sigma = I$ it reduces to $A_1A_2 = 0$.
- Geometrically the two forms must project onto orthogonal subspaces - exactly the SST/SSE split in ANOVA.
- This is why the numerator and denominator of the ANOVA F are independent.

STEP-BY-STEP SOLUTION

1. For $Y \sim N_n(0, I)$ and symmetric A_1, A_2 , Craig's theorem states that $Y'A_1Y$ and $Y'A_2Y$ are independent if and only if $A_1A_2 = 0$.
2. Chi-squaredness of each form requires each A_i to be idempotent; independence requires additionally the product to vanish.
3. Hence $A_1A_2 = 0$ - option (b).

Analysis of Variance

37 authentic PYQs in this topic (2018-2026)

Two-way Classification and ANOVA

LM

2018 Q6

Numerical

Easy

Q1.

For a two-way ANOVA with 6 rows, 4 columns and 3 observations per cell, the degrees of freedom for interaction and residual are respectively

A. 15 and 45

B. 15 and 48

C. 24 and 45

D. 24 and 48

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Interaction df $= (p-1)(q-1) = 5 \times 3 = 15$; error df $= pq(r-1) = 24 \times 2 = 48$.

TIPS & TRICKS

- Two-way with r observations per cell: α : $p-1$, β : $q-1$, interaction: $(p-1)(q-1)$, error: $pq(r-1)$, total: $pqr-1$.
- Check the arithmetic by adding: $5+3+15+48=71=pqr-1=72-1$. If your four df do not sum to $N-1$, you have made an error.
- The 45-vs-48 distractor comes from wrongly using $(pq-1)(r-1)$ or $pqr-pq-1$; always use $pq(r-1)$.

STEP-BY-STEP SOLUTION

- Here $p=6$ rows, $q=4$ columns, $r=3$ observations per cell, so $N=pqr=72$.
- Interaction df $= (p-1)(q-1) = (6-1)(4-1) = 15$.
- Error (residual) df $= pq(r-1) = 6 \times 4 \times 2 = 48$.
- Verification: $(p-1) + (q-1) + (p-1)(q-1) + pq(r-1) = 5 + 3 + 15 + 48 = 71 = N-1$. Correct.
- Hence 15 and 48 - option (b).

Q2.

Consider a two-way classification with r observations per cell and interaction present:

$$y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}; \quad i=1, \dots, p; j=1, \dots, q; k=1, \dots, r$$

The total number of parameters in this model is

A. $1+p+q+pq$

B. $p+q+pq$

C. pq

D. $p+q+1$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Count the symbols: one μ , p α 's, q β 's and pq γ 's, i.e. $1+p+q+pq$ (before any side conditions).

TIPS & TRICKS

- The question asks for the number of parameters, not the rank: side conditions such as $\sum \alpha_i = 0$ are not imposed here.
- The model is over-parametrised: $1+p+q+pq = (1+p)(1+q)$ parameters but only pq independent cell means, hence rank pq and $1+p+q$ redundancies.
- e_{ijk} is a random error, never counted as a parameter.

STEP-BY-STEP SOLUTION

1. The model is $y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$.
2. Parameters: the general mean μ (1); the row effects $\alpha_1, \dots, \alpha_p$ (p); the column effects $\beta_1, \dots, \beta_q$ (q); the interactions γ_{ij} (pq).
3. Total = $1+p+q+pq$.
4. Hence option (a). (Note this equals $(1+p)(1+q)$, while the rank of the design is only pq - the model is over-parametrised.)

SHARED CONTEXT

Three instructors each taught 12 students (randomly assigned to 3 teaching methods) the same material. Complete this two-way ANOVA table with interaction:

Source	df	Sum of squares (SS)	Mean SS	F-ratio
Methods	–	162	–	–
Instructors	–	90	–	–
Interaction	–	600	–	–
Error	–	900	–	–

Q3.

The degrees of freedom for methods, instructors, interaction and errors are respectively

- A. 2, 2, 95 and 107
- B. 3, 3, 101 and 107
- C. 2, 2, 4 and 27
- D. None of the above

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

3 methods $\times$ 3 instructors = 9 cells and $3 \times 12 = 36$ students, so $r=4$ per cell: $df = 2, 2, 4$ and $pq(r-1) = 9 \times 3 = 27$.

TIPS & TRICKS

- Two-way with replication: $p-1$, $q-1$, $(p-1)(q-1)$, $pq(r-1)$; they must sum to $pqr-1$.
- Check: $2+2+4+27=35=36-1$. The check catches every arithmetic slip.
- Options (a) and (b) treat the 36 students as 36 cells - a classic misreading of a replicated design.

STEP-BY-STEP SOLUTION

- Three instructors each taught 12 students, so $N=36$ observations, spread over $p=3$ methods and $q=3$ instructors, i.e. 9 cells with $r=36/9=4$ observations each.
- $df(\text{methods}) = p-1=2$; $df(\text{instructors}) = q-1=2$.
- $df(\text{interaction}) = (p-1)(q-1)=4$; $df(\text{error}) = pq(r-1)=9 \times 3=27$.
- Total $= 2+2+4+27=35=N-1$. Hence 2,2,4,27 - option (c).

SHARED CONTEXT

Three instructors each taught 12 students (randomly assigned to 3 teaching methods) the same material. Complete this two-way ANOVA table with interaction:

Source	df	Sum of squares (SS)	Mean SS	F-ratio
Methods	—	162	—	—
Instructors	—	90	—	—
Interaction	—	600	—	—
Error	—	900	—	—

Q4.

The mean sum of squares due to methods, instructors, interaction and errors are respectively

A. 81, 45, 150, 33.33

B. 81, 45, 6.5, 8.41

C. 54, 30, 5.94, 8.41

D. None of the above

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Divide each SS by its df from the previous item: $162/2=81$, $90/2=45$, $600/4=150$, $900/27=33.33$.

TIPS & TRICKS

- Mean square = SS/df , always. Get the df right first and this step is mechanical.
- $900/27=33.33$ is the error mean square - it becomes the denominator of every F in a fixed-effects model.
- Option (b)'s 6.5 and (c)'s 5.94 come from dividing by the wrong df; sanity-check that $MS(\text{interaction}) = 600/4=150$.

STEP-BY-STEP SOLUTION

1. From the df found above: methods 2, instructors 2, interaction 4, error 27.
2. $MS(\text{methods}) = 162/2=81$.
3. $MS(\text{instructors}) = 90/2=45$.
4. $MS(\text{interaction}) = 600/4=150$; $MS(\text{error}) = 900/27=33.33$.
5. Hence 81,45,150,33.33 - option (a).

SHARED CONTEXT

Three instructors each taught 12 students (randomly assigned to 3 teaching methods) the same material. Complete this two-way ANOVA table with interaction:

Source	df	Sum of squares (SS)	Mean SS	F-ratio
Methods	–	162	–	–
Instructors	–	90	–	–
Interaction	–	600	–	–
Error	–	900	–	–

Q5.

The F ratios pertaining to methods, instructors and interaction are, respectively

- A. 9.63, 5.35, 0.77
- B. 2.43, 1.35, 4.50
- C. 6.42, 3.57, 0.71
- D. None of the above

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Divide each mean square by $MSE = 33.33$: $81/33.33=2.43$, $45/33.33=1.35$, $150/33.33=4.50$.

TIPS & TRICKS

- In a fixed-effects two-way model every F uses MSE as denominator. (In a random-effects model the main effects would use $MS(\text{interaction})$ instead.)
- Quick check: the interaction F must be the largest here because $MS(\text{interaction}) = 150$ is the largest mean square.
- Option (a) divides by 8.41 and option (c) by a wrong MSE - always recompute $MSE = SSE/df$.

STEP-BY-STEP SOLUTION

1. $MSE = 900/27 = 33.33$.
2. $F(\text{methods}) = 81/33.33 = 2.43$.
3. $F(\text{instructors}) = 45/33.33 = 1.35$.
4. $F(\text{interaction}) = 150/33.33 = 4.50$.
5. Hence 2.43, 1.35, 4.50 - option (b).

Q6.

Two-way classification, one observation per cell, interaction present: $y_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ij}$, $i = 1, \dots, 5$; $j = 1, \dots, 4$.

The df for error SS are

- A. 4
- B. 5
- C. 12
- D. 0

Correct answer: D

EXAM SHORTCUT AI-derived explanation

With one observation per cell, interaction uses up all $(p-1)(q-1)$ residual df, leaving $pq(r-1) = pq(1-1) = 0$ for error.

TIPS & TRICKS

- Two-way, one observation per cell: either fit interaction and have no error df, or assume additivity and use $(p-1)(q-1)$ as error df.
- This is exactly why replication is needed to test interaction, and why Tukey's one-df test for non-additivity exists.
- Count check: $pq-1 = 19 = (5-1) + (4-1) + (4)(3) + 0 = 4 + 3 + 12 + 0$.

STEP-BY-STEP SOLUTION

1. Here $p=5$, $q=4$, $r=1$, so $N=pq=20$ and total df = 19.
2. Main effects take $p-1=4$ and $q-1=3$ df; interaction takes $(p-1)(q-1)=12$ df.
3. Error df = $pq(r-1) = 20 \times 0 = 0$; equivalently $19 - 4 - 3 - 12 = 0$.
4. Hence the error df is 0 - option (d). Interaction and error are completely confounded.

Q7.

Consider two-way classified data with 5 levels of factor A, 6 levels of factor B and 4 observations per cell. The degrees of freedom for error are:

- A. 120
- B. 100
- C. 96
- D. 90

Correct answer: D

EXAM SHORTCUT AI-derived explanation

Error df = $pq(r-1) = 5 \times 6 \times 3 = 90$.

TIPS & TRICKS

- $pq(r-1)$ - count cells, then subtract one df per cell for the cell mean.
- Check: $4 + 5 + 20 + 90 = 119 = pqr - 1 = 120 - 1$.
- Distractor 120 is pqr itself; 96 and 100 come from mis-multiplying $(pq-1)(r-1)$ style formulae.

STEP-BY-STEP SOLUTION

1. $p=5$, $q=6$, $r=4$ so $N=pqr=120$.
2. $\text{df}(A) = 4$, $\text{df}(B) = 5$, $\text{df}(AB) = 4 \times 5 = 20$.
3. $\text{df}(\text{error}) = pq(r-1) = 30 \times 3 = 90$.
4. Check: $4 + 5 + 20 + 90 = 119 = N - 1$. Option (d).

SHARED CONTEXT

The moisture content of three types of cheese made by two methods was recorded. Two pieces of cheese were measured for each type and for each method. Let the method be designated as factor A and type as factor B. The sum of squares for factor A, factor B and error are respectively 0.114075, 25.900117 and 0.661950.

Q8.

The mean squares for method and error are respectively:

- A. ~~0.114075~~ and 0.110325
- B. 0.057037 and 0.110325
- C. 0.114075 and 0.08274375
- D. 0.057037 and 0.08274375

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$a=2$ methods, $b=3$ types, $r=2$ per cell: $df(A) = 1$, $df(error) = ab(r-1)=6$. So $MSA = 0.114075/1$ and $MSE = 0.661950/6 = 0.110325$.

TIPS & TRICKS

- Read the layout from the words: 'two methods', 'three types', 'two pieces per type and method' gives $2 \times 3 \times 2 = 12$ observations.
- $df(A) = 1$ means MSA equals SSA numerically - the 0.057037 distractor wrongly divides by 2.
- The companion item (F-ratios) confirms the choice: only $MSE = 0.110325$ reproduces the printed F values.

STEP-BY-STEP SOLUTION

1. Factor A (method) has $a=2$ levels, factor B (type) has $b=3$ levels, with $r=2$ observations per cell, so $N=12$ and total $df = 11$.
2. $df(A) = a-1=1$; $df(B) = b-1=2$; $df(AB) = 2$; $df(error) = ab(r-1)=6$.
3. $MS(\text{method}) = \frac{0.114075}{1} = 0.114075$.
4. $MS(\text{error}) = \frac{0.661950}{6} = 0.110325$.
5. Hence option (a).

SHARED CONTEXT

The moisture content of three types of cheese made by two methods was recorded. Two pieces of cheese were measured for each type and for each method. Let the method be designated as factor A and type as factor B. The sum of squares for factor A, factor B and error are respectively 0.114075, 25.900117 and 0.661950.

Q9.

The values of F-statistic for factors A and B are respectively:

- A. 1.034 and 78.2539
- B. 0.51699 and 117.381
- C. **1.034 and 117.381**
- D. 0.51699 and 78.2539

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$$F_A = 0.114075 / 0.110325 = 1.034 \text{ and } F_B = (25.900117 / 2) / 0.110325 = 117.381.$$

TIPS & TRICKS

- Always compute $MSB = SSB/df(B)$ first: $25.900117/2 = 12.9500585$.
- $F_A \approx 1$ means the method effect is negligible; F_B is enormous, so cheese type dominates - a useful sanity read.
- Distractors halve F_A (wrong df for A) or use the wrong MSE.

STEP-BY-STEP SOLUTION

1. From the previous item: $MSA = 0.114075$, $MSE = 0.110325$, and $MSB = \frac{25.900117}{2} = 12.9500585$.

2. $F_A = \frac{MSA}{MSE} = \frac{0.114075}{0.110325} = 1.034$.

3. $F_B = \frac{MSB}{MSE} = \frac{12.9500585}{0.110325} = 117.381$.

4. Hence 1.034 and 117.381 - option (c).

Q10.

Consider a two-way classification with r observations per cell $Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$, where $i = 1, 2, \dots, p$; $j = 1, 2, \dots, q$; $k = 1, 2, \dots, r$. Consider the ANOVA table for this classification:

Source	Degrees of freedom
Due to α	t
Due to β	u
Interaction	v
Error	w
Total	$pqr - 1$

What are the values of t , u , v and w respectively?

A. $p-1, q-1, (p-1)(q-1), pq(r-1)$

B. $q-1, p-1, (p-1)(q-1), pq(r-1)$

C. $r-1, q-1, (r-1)(q-1), pq(r-1)$

D. $p-1, r-1, (r-1)(p-1), pq(r-1)$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$t=p-1$, $u=q-1$, $v=(p-1)(q-1)$, $w=pq(r-1)$ - the standard two-way-with-replication table.

TIPS & TRICKS

- Match each df to its own index range: α runs over p levels, so $p-1$; β over q , so $q-1$.
- Options (b), (c), (d) scramble p , q and r ; check the first entry and the rest follows.
- Verify by summation: $(p-1) + (q-1) + (p-1)(q-1) + pq(r-1) = pqr - 1$.

STEP-BY-STEP SOLUTION

1. α_i has p levels subject to one constraint, so $t=p-1$.
2. β_j has q levels subject to one constraint, so $u=q-1$.
3. Interaction: $v=(p-1)(q-1)$.
4. Error: $w=pq(r-1)$, one df lost per cell out of the r replicates.
5. These sum to $pqr-1$, matching the total. Option (a).

Q11.

In a Randomized Block Design with 4 blocks and 5 treatments having one missing observation, the error degrees of freedom are:

- A. 12
- B. 11**
- C. 10
- D. 9

Correct answer: B

EXAM SHORTCUT AI-derived explanation

RBD error df $= (b-1)(v-1) = 3 \times 4 = 12$; each missing value costs one more df, so $12-1 = 11$.

TIPS & TRICKS

- Missing-plot technique: estimating a missing value uses up one error degree of freedom.
- Generic rule: error df $= (b-1)(v-1) - (\text{number of missing observations})$.
- The full-data answer 12 is planted as option (a) to catch candidates who forget the adjustment.

STEP-BY-STEP SOLUTION

1. For a randomized block design with $b=4$ blocks and $v=5$ treatments, error df $= (b-1)(v-1) = 3 \times 4 = 12$.
2. One observation is missing and is estimated by the missing-plot formula, which consumes one degree of freedom from the error line.
3. Error df $= 12-1 = 11$.
4. Hence option (b).

Q12.

Three different methods of analysis are used to determine the amount of a certain constituent in a sample. Five different analysts generate one observation each under each of these methods. Various sums of squares are obtained as:

$$TSS=97.6, \quad SSA \text{ (Analysts)}=4.27, \quad SSB \text{ (Methods)}=79.6$$

What is the value of MSE, the Mean Sum of Squares due to error?

A. 2.89

B. 1.96

C. **1.72**

D. 1.53

Correct answer: C

EXAM SHORTCUT AI-derived explanation

$N=15$, total df 14; error df $=14-4-2=8$. $SSE=97.6-4.27-79.6=13.73$, so $MSE = 13.73/8=1.72$.

TIPS & TRICKS

- Two-way with one observation per cell: error df $=(p-1)(q-1)=(3-1)(5-1)=8$.
- SSE is always TSS minus the two main-effect sums of squares when there is no replication.
- Sanity check: MSE should be the smallest mean square here; $79.6/2=39.8$ for methods dwarfs it, so methods matter.

STEP-BY-STEP SOLUTION

1. There are $p=3$ methods and $q=5$ analysts with one observation per cell, so $N=15$ and total df $=14$.
2. $df(\text{analysts}) = q-1=4$; $df(\text{methods}) = p-1=2$; $df(\text{error}) = (p-1)(q-1)=8$.
3. $SSE=TSS-SSA-SSB=97.6-4.27-79.6=13.73$.
4. $MSE = \frac{13.73}{8} = 1.716 \approx 1.72$.
5. Hence option (c).

Q13.

For a two-way classification with one observation per cell, if p and q are the levels of the two factors A and B, then the degrees of freedom for error are:

- A. $pq-1$
- B. $p(q-1)$
- C. $q(p-1)$
- D. $(p-1)(q-1)$

Correct answer: D

EXAM SHORTCUT AI-derived explanation

With $r=1$ and no interaction term, error df $= (p-1)(q-1)$.

TIPS & TRICKS

- Total df $pq-1$ minus $(p-1)$ minus $(q-1)$ leaves $(p-1)(q-1)$ - do the subtraction rather than memorising.
- If an interaction term is fitted with one observation per cell, error df collapses to 0 (asked in 2020 Q49).
- For a randomized block design with b blocks and v treatments this same formula reads $(b-1)(v-1)$.

STEP-BY-STEP SOLUTION

1. With one observation per cell the total number of observations is pq , so total df $= pq-1$.
2. Factor A takes $p-1$ df and factor B takes $q-1$ df.
3. Error df $= (pq-1) - (p-1) - (q-1) = pq - p - q + 1 = (p-1)(q-1)$.
4. Hence option (d).

Q14.

In the analysis of two-way classified data with m observations per cell, where the factor A is at p levels and the factor B is at q levels, the degrees of freedom for the sum of squares due to error are:

- A. $pq(m-1)$
- B. $m(pq-1)$
- C. $m(p-1)(q-1)$
- D. $(m-1)(p-1)(q-1)$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

Error df = $pq(m-1)$: one df lost per cell from the m replicates in each of the pq cells.

TIPS & TRICKS

- Think of the error line as pooling pq within-cell sums of squares, each with $m-1$ df.
- Check by addition: $(p-1)+(q-1)+(p-1)(q-1)+pq(m-1)=pqm-1$.
- If $m=1$ this gives 0, which is why interaction cannot be tested without replication.

STEP-BY-STEP SOLUTION

1. Total observations = pqm , so total df = $pqm-1$.
2. Main effects take $p-1$ and $q-1$ df; interaction takes $(p-1)(q-1)$.
3. Error df = $(pqm-1)-(p-1)-(q-1)-(p-1)(q-1)=pqm-pq=pq(m-1)$.
4. Hence option (a).

Q15.

In the following Analysis of Variance Table, few entries are missing.

Source of variation	Degrees of freedom	Sum of squares	Variance ratio
Treatment	4	16.4	2.05
Variety	x	28	7
Error	y	z	
Total	15		

What are the values of x , y and z respectively?

- A. 4,9,18
- B. 2,9,16
- C. 9,2,18
- D. **2,9,18**

Correct answer: D

EXAM SHORTCUT AI-derived explanation

$MST = 16.4/4 = 4.1$ and $F = 2.05$ give $MSE = 2$. Then $MSV = 7 \times 2 = 14$ so $x = 28/14 = 2$; $y = 15 - 4 - 2 = 9$; $z = 2 \times 9 = 18$.

TIPS & TRICKS

- Work backwards from a printed F : $MSE = MS(\text{source})/F$. This unlocks every remaining cell.
- Degrees of freedom must add up: $4 + x + y = 15$.
- Final check: z/y must return the MSE you started with - here $18/9 = 2$.

STEP-BY-STEP SOLUTION

1. Treatment: $MS = 16.4/4 = 4.1$ and variance ratio 2.05, so $MSE = 4.1/2.05 = 2$.
2. Variety: variance ratio 7 means $MS(\text{variety}) = 7 \times MSE = 14$; since $SS = 28$, $x = 28/14 = 2$.
3. Total $df = 15$, so $y = 15 - 4 - 2 = 9$.
4. $z = SSE = MSE \times y = 2 \times 9 = 18$.
5. Hence $(x, y, z) = (2, 9, 18)$ - option (d).

One-way Classification and ANOVA

Q1.

As the variability due to chance in an analysis of variance decreases, the F -statistic in the ANOVA

- A. will always increase**
- B. will always decrease
- C. will remain constant
- D. requires additional information to claim any change

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

$F = \text{MST} / \text{MSE}$ and chance variability is exactly MSE, the denominator. Shrink the denominator and F rises.

TIPS & TRICKS

- 'Variability due to chance' = error/residual variation = MSE.
- In ANOVA the signal-to-noise reading is literal: F = between-group variance / within-group variance.
- Smaller MSE also means a more sensitive experiment - this is why blocking and replication are used.

STEP-BY-STEP SOLUTION

1. The ANOVA F -statistic is $F = \frac{\text{MST}}{\text{MSE}} = \frac{\text{between-treatment mean square}}{\text{error mean square}}$.
2. 'Variability due to chance' is the error (residual) component, which is the denominator MSE.
3. With the treatment mean square held fixed, decreasing MSE increases the ratio.
4. Hence the F -statistic will always increase - option (a).

Q2.

When k population means in an ANOVA are truly different from each other, it is likely that the average error deviation, for a significant F -statistic,

- A. is relatively large compared to the average treatment deviations
- B. is relatively small compared to the average treatment deviations**
- C. is equal to the average treatment deviations
- D. differs significantly between at least two of the populations

Correct answer: B

EXAM SHORTCUT AI-derived explanation

A significant F means treatment deviations dominate error deviations, so the average error deviation is relatively small.

TIPS & TRICKS

- F large $\iff$ MST $\gg$ MSE $\iff$ error deviations small relative to treatment deviations.
- Option (d) is a category error: the error deviation is a single pooled quantity, it does not 'differ between populations'.
- Remember the logic of rejection: reject H_0 when between-group spread cannot be explained by within-group noise.

STEP-BY-STEP SOLUTION

1. When the k population means genuinely differ, the treatment (between-group) sum of squares is inflated by the real differences among means.
2. A significant $F = \text{MST}/\text{MSE}$ requires the numerator to be large relative to the denominator.
3. Equivalently the average error (within-group) deviation must be relatively small compared with the average treatment deviations.
4. Hence option (b).

Q3.

Consider the ANOVA table:

Source	df	Sum of squares	Mean SS	Test statistic
Treatment	w	x	y	5.22
Error	12	z	25	
Total	14	561		

The values of w, x, y, z are respectively

A. 2, 261, 130.5, 300

B. 2, 300, 150, 261

C. 2, 250, 125, 311

D. 2, 311, 155.5, 250

Correct answer: A

EXAM SHORTCUT[AI-derived explanation](#)

$z = 12 \times 25 = 300$; $x = 561 - 300 = 261$; $w = 14 - 12 = 2$; $y = 261/2 = 130.5$. Check $130.5/25 = 5.22$.

TIPS & TRICKS

- Order of filling an ANOVA table: df by subtraction, then SS by subtraction, then $MS = SS/df$, then $F = MS/MSE$.
- Always verify with the printed F : it is a free consistency check on all four unknowns.
- Total df = $N-1$, so $N=15$ observations and $k-1=2$ means $k=3$ treatments.

STEP-BY-STEP SOLUTION

1. Error: df = 12 and MSS = 25, so $z = SSE = 12 \times 25 = 300$.
2. Total SS = 561, hence treatment SS $x = 561 - 300 = 261$.
3. Treatment df $w = \text{total df} - \text{error df} = 14 - 12 = 2$.
4. Treatment mean SS $y = x/w = 261/2 = 130.5$.
5. Check: $F = y/25 = 130.5/25 = 5.22$, matching the table. Hence $(w, x, y, z) = (2, 261, 130.5, 300)$ - option (a).

SHARED CONTEXT

An experiment tested the difference in mean weights of a fish species caught in three lakes, at significance level 0.05. Incomplete ANOVA table:

Source	df	Sum of squares	Mean SS	F statistic
Treatment	–	17.04	–	–
Error	9	–	–	–
Total	11	35.04	–	–

Q4.

The null hypothesis for the analysis is

- A. $H_0: \mu_1 = \mu_2 = \dots = \mu_9 (= \mu)$
- B. $H_0: \mu_1 = \mu_2 = \mu_3 = 0$
- C. ~~$H_0: \mu_1 = \mu_2 = \mu_3 (= \mu)$~~
- D. at least one pair of fish populations have same mean

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Three lakes = three treatments, so $H_0: \mu_1 = \mu_2 = \mu_3$. The '9' in option (a) is the error df, not a number of groups.

TIPS & TRICKS

- In one-way ANOVA the null is always equality of all treatment means, never equality to zero.
- Total df = 11 means $N=12$ observations; treatment df = $11-9=2$ means $k=3$ groups.
- Option (d) states the alternative badly - 'at least one pair has the same mean' is not the negation of H_0 .

STEP-BY-STEP SOLUTION

1. Total df = $N-1=11$ so $N=12$; error df = $N-k=9$ so $k=3$ groups, matching the three lakes.
2. One-way ANOVA tests equality of the k treatment means.
3. Hence $H_0: \mu_1 = \mu_2 = \mu_3 (= \mu)$ - option (c).

SHARED CONTEXT

An experiment tested the difference in mean weights of a fish species caught in three lakes, at significance level 0.05. Incomplete ANOVA table:

Source	df	Sum of squares	Mean SS	F statistic
Treatment	–	17.04	–	–
Error	9	–	–	–
Total	11	35.04	–	–

Q5.

If the critical value of F at 5% level is 4.2565, the appropriate interpretation is

- A. Reject H_0 ; all populations of fish have different mean weights.
- B. Reject H_0 ; exactly two populations of fish have different mean weights.
- C. **Reject H_0 ; the mean weight of at least one fish population differs from others.**
- D. There is insufficient evidence to claim that the mean weights differ.

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

$F = 8.52/2 = 4.26 > 4.2565$, so reject H_0 . Rejection only says at least one mean differs - never that all differ or exactly two differ.

TIPS & TRICKS

- Rejecting the ANOVA F licences only the statement 'not all means are equal'. Identifying which pairs differ requires Tukey/Scheffe/SNK.
- This is a knife-edge decision (4.26 against 4.2565) - compute the F exactly rather than rounding early.
- Options claiming 'all populations differ' or 'exactly two differ' are always wrong after a global F -test.

STEP-BY-STEP SOLUTION

- Treatment df = $11 - 9 = 2$, so $MST = 17.04/2 = 8.52$.
- $SSE = 35.04 - 17.04 = 18.00$, so $MSE = 18/9 = 2$.
- $F = 8.52/2 = 4.26$, and the 5% critical value is 4.2565. Since $4.26 > 4.2565$, reject H_0 .
- The correct interpretation of rejection is that at least one population mean differs from the others - option (c).

SHARED CONTEXT

An experiment tested the difference in mean weights of a fish species caught in three lakes, at significance level 0.05. Incomplete ANOVA table:

Source	df	Sum of squares	Mean SS	F statistic
Treatment	–	17.04	–	–
Error	9	–	–	
Total	11	35.04		

Q6.

The value of the F statistic in the table equals

- A. 2.46
- B. 4.26
- C. 6.24
- D. 8.52

Correct answer: B

EXAM SHORTCUT AI-derived explanation

$MST = 17.04/2 = 8.52$; $MSE = (35.04 - 17.04)/9 = 2$; $F = 8.52/2 = 4.26$.

TIPS & TRICKS

- Fill the table in the fixed order $df \rightarrow SS \rightarrow MS \rightarrow F$; each step is a one-line subtraction or division.
- Treatment $df = \text{total } df - \text{error } df = 11 - 9 = 2$.
- Distractor 8.52 is MST itself and 2.46 inverts the ratio - always put the treatment mean square on top.

STEP-BY-STEP SOLUTION

- Treatment $df = 11 - 9 = 2$; $SST = 17.04$, so $MST = 17.04/2 = 8.52$.
- $SSE = SS(\text{Total}) - SST = 35.04 - 17.04 = 18.00$; error $df = 9$, so $MSE = 18.00/9 = 2.00$.
- $F = \frac{MST}{MSE} = \frac{8.52}{2.00} = 4.26$.
- Hence option (b).

SHARED CONTEXT

An experiment tested the difference in mean weights of a fish species caught in three lakes, at significance level 0.05. Incomplete ANOVA table:

Source	df	Sum of squares	Mean SS	F statistic
Treatment	–	17.04	–	–
Error	9	–	–	–
Total	11	35.04	–	–

Q7.

If all observations from three lakes are pooled into a single group, they would have a standard deviation equal to

A. 2.000

B. 1.414

C. 4.128

D. 5.919

Correct answer: B

EXAM SHORTCUT AI-derived explanation

Pooling the three groups means pooling their within-group variability: $s_p^2 = \text{MSE} = 2$, so $s_p = \sqrt{2} = 1.414$.

TIPS & TRICKS

- The pooled standard deviation in ANOVA is always $\sqrt{\text{MSE}}$ - MSE is by construction the pooled within-group variance.
- Distractors here are $\text{MSE} = 2$ itself, $\sqrt{\text{SST}} = 4.128$ and $\sqrt{\text{SS(Total)}} = 5.919$; recognise each and discard.
- Do not confuse the pooled (within-group) SD with the SD of all 12 values treated as one sample, $\sqrt{35.04/11} = 1.785$, which is not offered.

STEP-BY-STEP SOLUTION

1. $\text{SSE} = 35.04 - 17.04 = 18.00$ with error $\text{df} = 9$, so $\text{MSE} = 2.00$.

2. MSE is the pooled estimate of the common within-group variance σ^2 : $s_p^2 = \frac{\sum_i (n_i - 1)s_i^2}{\sum_i (n_i - 1)} = \text{MSE}$.

3. Hence the pooled standard deviation is $s_p = \sqrt{2} = 1.414$.

4. Option (b).

LM

2021 Q8

Numerical

Easy

Q8.

Five methods of packing frozen foods were compared taking six observations per method (response: ascorbic acid, mg/100 g). The df for error sum of squares are

- A. 26
- B. 25
- C. 24
- D. 20

Correct answer: B

EXAM SHORTCUT

AI-derived explanation

One-way with $k=5$ methods and $n=6$ each: $N=30$, error df $=N-k=25$ (equivalently $k(n-1)=5\times 5$).

TIPS & TRICKS

- Balanced one-way: error df $=k(n-1)$, treatment df $=k-1$, total df $=kn-1$.
- Check the three add up: $4+25=29=30-1$.
- Distractor 24 comes from $kn-k-1$, distractor 26 from $N-k+1$ - both fail the addition check.

STEP-BY-STEP SOLUTION

1. There are $k=5$ treatments (packing methods) with $n=6$ observations each, so $N=30$.
2. Treatment df $=k-1=4$; total df $=N-1=29$.
3. Error df $=N-k=30-5=25$, i.e. $k(n-1)=5\times 5=25$.
4. Hence option (b).

Q9.

Three methods A, B and C of packing frozen foods were compared. The response variable was ascorbic acid (mg/100 g). The data was collected by taking seven observations for each of the methods. The sum of squares due to treatments and that due to total are 147.3456 and 202.3126 respectively. What is the value of F-statistic?

A. 24.1256

B. 24.0156

C. 23.5216

D. 23.1256

Correct answer: A

EXAM SHORTCUT AI-derived explanation

$k=3$, $N=21$: $MST = 147.3456/2 = 73.6728$; $SSE = 202.3126 - 147.3456 = 54.967$, $MSE = 54.967/18 = 3.05372$; $F = 24.1256$.

TIPS & TRICKS

- Fill df first: treatment $k-1=2$, error $N-k=18$, total 20.
- SSE is always obtained by subtraction when only SST and SS(Total) are given.
- Keep four decimals throughout - the four options differ only in the third decimal place.

STEP-BY-STEP SOLUTION

1. $k=3$ methods with 7 observations each, so $N=21$; treatment df = 2, error df = 18.

$$2. MST = \frac{147.3456}{2} = 73.6728.$$

$$3. SSE = 202.3126 - 147.3456 = 54.9670, \text{ so } MSE = \frac{54.9670}{18} = 3.05372.$$

$$4. F = \frac{73.6728}{3.05372} = 24.1256.$$

5. Hence option (a).

Q10.

Consider a one-way classification model $Y_{ij} = \mu + \alpha_i + e_{ij}$ where $i = 1, 2, 3, \dots, k$; $j = 1, 2, 3, \dots, n$ and $e_{ij} \sim N(0, \sigma^2)$. Assume that $N = \sum_{i=1}^k n_i$. We are testing the hypothesis $H_0: \alpha_1 = \alpha_2 = \dots = \alpha_k = \alpha$ (say). Consider the ANOVA table for this classification:

Source	Degrees of freedom
Treatment	u
Error	v
Total	$N-1$

What are the values of u and v respectively?

- A. $k, N-k-1$
- B. $k-2, N-k+1$
- C. $k-1, N-k$
- D. $k-3, N-k+2$

Correct answer: C

EXAM SHORTCUT AI-derived explanation

One-way classification: treatment df = $k-1$, error df = $N-k$.

TIPS & TRICKS

- The three df must satisfy $(k-1) + (N-k) = N-1$; only option (c) does.
- This addition check alone eliminates the other three options without any thought about the model.
- $N = \sum n_i$ allows unequal group sizes - the formula is unchanged.

STEP-BY-STEP SOLUTION

1. There are k treatments, so the treatment sum of squares carries $u = k-1$ degrees of freedom.
2. Total df = $N-1$.
3. Error df = $v = (N-1) - (k-1) = N-k$.
4. Hence $(u, v) = (k-1, N-k)$ - option (c).

Q11.

Consider the following ANOVA table:

Sources of Variation	Degrees of freedom	Sum of squares	Mean sum of squares	Test statistic
Treatments	m	p	r	5.22
Error	12	q	25	
Total	14	561		

What are the values of p , q and r respectively?

A. 261, 300, 130.5

B. 300, 261, 130.5

C. 250, 300, 128.5

D. 261, 300, 128.5

Correct answer: A

EXAM SHORTCUT

AI-derived explanation

$q = \text{SSE} = 12 \times 25 = 300$; $p = 561 - 300 = 261$; $r = 261/2 = 130.5$ (and $130.5/25 = 5.22$ confirms).

TIPS & TRICKS

- This is a verbatim numerical repeat of 2019 Q5 with the labels changed - same 5.22, 12, 25, 14, 561.
- Watch the order in which the question asks for the symbols: here it is p (SS), q (SS), r (MS), not w, x, y, z .
- The printed F is a free check - if MST/MSE does not reproduce it, re-do the arithmetic.

STEP-BY-STEP SOLUTION

1. Error line: $\text{df} = 12$, $\text{MS} = 25$, so $q = \text{SSE} = 12 \times 25 = 300$.
2. Total $\text{SS} = 561$, so treatment $\text{SS } p = 561 - 300 = 261$.
3. Treatment $\text{df} = 14 - 12 = 2$, so $r = \text{MST} = 261/2 = 130.5$.
4. Check: $F = 130.5/25 = 5.22$ as printed. Hence $(p, q, r) = (261, 300, 130.5)$ - option (a).

Q12.

Which of the following statements are correct regarding ANOVA technique?

- I. It is used to test the equality of several population means.
- II. It is used to test the equality of several population variances.
- III. Its origin lies in agricultural experiments.

Select the answer using the code given below.

- A. I and II only
- B. II and III only
- C. I and III only**
- D. I, II and III

Correct answer: C

EXAM SHORTCUT AI-derived explanation

ANOVA tests equality of means (I true) and was developed by R. A. Fisher at Rothamsted agricultural station (III true); it does not test equality of variances (II false).

TIPS & TRICKS

- ANOVA compares means by *analysing variances* - the technique uses variances but the hypothesis is about means.
- Equality of several variances is tested by Bartlett's or Levene's test, not ANOVA.
- Homogeneity of variances is an *assumption* of ANOVA, not its hypothesis - that inversion is the trap.

STEP-BY-STEP SOLUTION

1. Statement I: one-way ANOVA tests $H_0: \mu_1 = \dots = \mu_k$. True.
2. Statement II: tests for equality of several variances are Bartlett's test, Levene's test and Hartley's $F_{\max}$ - not ANOVA. False.
3. Statement III: the technique originated in R. A. Fisher's agricultural field experiments at Rothamsted. True.
4. Hence I and III only - option (c).

Q13.

In ANOVA for one-way classified data, which of the following statements are correct?

I. The value of various sum of squares is independent of change of origin.

II. The value of various sum of squares is independent of change of scale.

III. The value of variance ratio is independent of change of origin.

IV. The value of variance ratio is independent of change of scale.

Select the answer using the code given below.

A. I, II and III

B. I, II and IV

C. I, III and IV

D. II, III and IV

Correct answer: C

EXAM SHORTCUT

AI-derived explanation

Sums of squares are invariant to change of origin but multiply by c^2 under change of scale; the F -ratio, being a ratio of two such quantities, is invariant to both.

TIPS & TRICKS

- Change of origin $y \rightarrow y+a$: all deviations $y_i - \bar{y}$ unchanged, so every SS is unchanged.
- Change of scale $y \rightarrow cy$: every SS multiplies by c^2 , so SS is NOT scale-invariant - statement II is the false one.
- In the ratio $F = \text{MST}/\text{MSE}$ the factor c^2 cancels, making F invariant to both transformations.

STEP-BY-STEP SOLUTION

1. Statement I: replacing y_i by $y_i + a$ leaves every deviation from a mean unchanged, so all sums of squares are unchanged. True.
2. Statement II: replacing y_i by cy_i multiplies every sum of squares by c^2 , so sums of squares are not scale-invariant. False.
3. Statement III: since each SS is origin-invariant, so is the variance ratio. True.
4. Statement IV: under scaling both MST and MSE gain the same factor c^2 , which cancels in the ratio. True.
5. Hence I, III and IV - option (c).

Q14.

For the validity of F -test in ANOVA, which of the following are assumed?

- I. The observations are independent.
- II. Parent population from which observations are taken is normally distributed.
- III. The variances of the population from which observations are taken may differ from one another.
- IV. Different effects are additive in nature.

Select the correct answer using the code given below.

- A. I, II and III
- B. I, II and IV**
- C. I, III and IV
- D. II, III and IV

Correct answer: B

EXAM SHORTCUT AI-derived explanation

The assumptions are independence, normality, homoscedasticity and additivity. Statement III says variances 'may differ', which contradicts homoscedasticity, so it is out: I, II and IV.

TIPS & TRICKS

- Four assumptions of the ANOVA F -test: independent observations, normal parent population, equal variances, additive effects.
- Any option asserting unequal variances is automatically wrong - variance homogeneity is non-negotiable.
- Violation remedies: transformations (log, square-root, arcsine) restore additivity and stabilise variance.

STEP-BY-STEP SOLUTION

1. Statement I: independence of observations is required. True.
2. Statement II: normality of the parent population is required for the F distribution to hold exactly. True.
3. Statement III: the F -test requires homoscedasticity - equal variances. Saying they 'may differ' contradicts the assumption. False.
4. Statement IV: the effects of the various factors must be additive. True.
5. Hence I, II and IV - option (b).

Multiple Comparison Tests (Tukey, Scheffe, SNK-Duncan)

Q1.

Consider the following statements:

1. Scheffe's test of multiple comparisons tests all simple and complex contrasts.
2. Bartlett's test of multiple comparisons tests simple contrasts.
3. Tukey's test of multiple comparisons reduces the Type-I error in the test.

Codes:

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Scheffe covers all contrasts (1 true); Bartlett is a variance-homogeneity test, not a multiple-comparison procedure (2 false); Tukey controls family-wise Type-I error (3 true). So 1 and 3 only.

TIPS & TRICKS

- Bartlett's test = homogeneity of variances. Planting it among multiple-comparison procedures is a recurring UPSC trap.
- Scheffe: all simple and complex contrasts, most conservative. Tukey: all pairwise comparisons via the studentised range. SNK/Duncan: stepwise on ordered means.
- Every genuine multiple-comparison procedure exists to control the inflated family-wise Type-I error.

STEP-BY-STEP SOLUTION

1. Statement 1: Scheffe's method gives simultaneous confidence bounds for every contrast - simple (pairwise) and complex - so it is true.
2. Statement 2: Bartlett's test tests $H_0: \sigma_1^2 = \dots = \sigma_k^2$, i.e. homogeneity of variances. It is not a multiple-comparison test at all, so the statement is false.
3. Statement 3: Tukey's HSD uses the studentised range distribution to hold the family-wise error rate at α , which reduces the Type-I error inflation caused by repeated t -tests. True.
4. Hence 1 and 3 only - option (c).

Q2.

Consider the following statements for multiple comparison tests conducted after a significant result is obtained in ANOVA:

1. The t -test on all pairwise comparisons inflates Type-I error.
2. Scheffe's test requires equal group sizes for multiple comparisons.
3. Newman-Keuls test reduces the risk of Type-II error.

Codes:

- A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

Correct answer: C

EXAM SHORTCUT AI-derived explanation

1 is the standard motivation for multiple comparisons (true); Scheffe does not need equal group sizes (2 false); SNK is stepwise and more powerful, hence lower Type-II risk (3 true). So 1 and 3 only.

TIPS & TRICKS

- Family-wise error with m independent tests at level α is $1-(1-\alpha)^m$ - it explodes with m ; this is why all-pairwise t -testing is wrong.
- Scheffe is the most flexible procedure: unequal n_i are allowed. It is Tukey's HSD that is designed for (and works best with) equal n .
- Conservativeness ladder: Scheffe > Tukey > SNK > Duncan. Less conservative = more power = smaller Type-II error.

STEP-BY-STEP SOLUTION

1. Statement 1: performing all $\binom{k}{2}$ pairwise t -tests at level α inflates the family-wise Type-I error far above α . True.
2. Statement 2: Scheffe's procedure is derived from the F -distribution and applies to all contrasts with arbitrary (possibly unequal) group sizes. The statement is false.
3. Statement 3: the Student-Newman-Keuls procedure applies progressively smaller critical ranges to ordered means, making it more powerful than Tukey and so reducing the Type-II error risk. True.
4. Hence 1 and 3 only - option (c).

Q3.

Consider the following statements:

1. Tukey's test of multiple comparisons reduces the Type-I error in the test.
2. Student-Newman-Keuls test uses a stepwise procedure.

Codes:

- A. 1 only
- B. 2 only
- C. Both 1 and 2**
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Tukey's HSD controls the family-wise Type-I error, and SNK is by construction a stepwise (step-down) procedure on ordered means. Both true.

TIPS & TRICKS

- Tukey uses the studentised range $q_{\alpha}(k, \nu)$ with one critical value for all pairs.
- SNK compares means ordered by magnitude and uses a critical range that shrinks with the number of means spanned - hence 'stepwise'.
- Ranking by conservativeness: Scheffe > Tukey > SNK > Duncan.

STEP-BY-STEP SOLUTION

1. Statement 1: Tukey's honestly significant difference procedure uses a single studentised-range critical value so that the probability of any false rejection in the whole family is held at α - it reduces the Type-I error inflation of repeated t -tests. True.
2. Statement 2: the Student-Newman-Keuls procedure orders the means and tests them in steps, using a critical range that depends on how many means are spanned. It is explicitly a stepwise multiple-range procedure. True.
3. Both correct - option (c).

Q4.

A manufacturing company wishes to determine whether one of the three machines A, B and C is faster than the others in producing a certain output. The mean hourly output of the three machines is respectively 32, 37 and 27. It is given that at 5% level of significance, the critical difference between any two means to be significant is 5.62. Which one of the following is correct?

- A. Machines A and B differ significantly
- B. Machines A and C differ significantly
- C. Machines B and C differ significantly
- D. No conclusion can be drawn

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Differences are $|32-37|=5$, $|32-27|=5$, $|37-27|=10$. Only 10 exceeds the critical difference 5.62, so B and C differ.

TIPS & TRICKS

- Compute all $\binom{3}{2}=3$ pairwise absolute differences and compare each with the single critical difference.
- The critical difference (least significant / honestly significant difference) is the same for every pair in a balanced design.
- Two differences of exactly 5 fall just below 5.62 - the examiner is testing whether you compare rather than eyeball.

STEP-BY-STEP SOLUTION

1. Mean outputs: $A=32$, $B=37$, $C=27$; critical difference = 5.62 at 5%.
2. $|A-B|=|32-37|=5 < 5.62$: not significant.
3. $|A-C|=|32-27|=5 < 5.62$: not significant.
4. $|B-C|=|37-27|=10 > 5.62$: significant.
5. Hence only machines B and C differ significantly - option (c).

Fixed, Random and Mixed Effects Models

Q1.

For GML model $E(y_i) = \alpha_{i1}\beta_1 + \dots + \alpha_{ip}\beta_p$, $V(y_i) = \sigma^2$, $\text{COV}(y_i, y_j) = 0$:

1. If $\alpha_1, \dots, \alpha_p$ are linearly independent ($p \leq n$), the model is of full rank.
2. If all β 's are fixed constants, it is called a fixed effect model.

Codes:

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2

Correct answer: C

EXAM SHORTCUT AI-derived explanation

Linearly independent columns with $p \leq n$ gives $\text{rank}(X) = p$, a full-rank model; and treating all β_j as unknown constants is precisely the fixed-effects model. Both true.

TIPS & TRICKS

- Full-rank model $\iff \text{rank}(X) = p \iff (X'X)^{-1}$ exists $\iff$ every $\lambda'\beta$ estimable.
- Fixed effects: levels are the entire population of interest. Random effects: levels are a random sample and β_j are random variables with variance σ^2_β . Mixed: one of each.
- In random-effects models inference targets variance components, not the individual effects.

STEP-BY-STEP SOLUTION

1. Statement 1: if the p column vectors $\alpha_1, \dots, \alpha_p$ are linearly independent and $p \leq n$, then $\text{rank}(X) = p$, which is the definition of a full-rank (non-singular) model. True.
2. Statement 2: when every β_j is an unknown but fixed constant, the model is a fixed-effect model. True.
3. Both correct - option (c).

Q2.

For a two-way classified random effect model with p levels of factor A and q levels of factor B having m observations per cell, the estimate of variance component σ_A^2 is given by:

A. $\frac{MSA - MS(AB)}{qm}$

B. $\frac{MSB - MS(AB)}{pm}$

C. $\frac{MSA - MSB}{pq}$

D. $\frac{MSA + MSB}{pq}$

Correct answer: A

EXAM SHORTCUT AI-derived explanation

From the expected mean squares of a two-way random model, $E(MSA) = \sigma^2 + m\sigma_{ab}^2 + qm\sigma_a^2$ and

$$E(MS_{AB}) = \sigma^2 + m\sigma_{ab}^2, \text{ so } \hat{\sigma}_a^2 = \frac{MSA - MS(AB)}{qm}.$$

TIPS & TRICKS

- Memorise the EMS table: $E(MSE) = \sigma^2$; $E(MS_{AB}) = \sigma^2 + m\sigma_{ab}^2$; $E(MSA) = \sigma^2 + m\sigma_{ab}^2 + qm\sigma_a^2$; $E(MSB) = \sigma^2 + m\sigma_{ab}^2 + pm\sigma_b^2$.
- The divisor is always 'number of observations at each level of that factor' = qm for A and pm for B.
- In a random model the main-effect F 's use MS_{AB} , not MSE , as denominator - the single most examinable asymmetry in this topic.

STEP-BY-STEP SOLUTION

1. For the two-way random effects model with m observations per cell the expected mean squares are

$$E(MSA) = \sigma^2 + m\sigma_{ab}^2 + qm\sigma_a^2 \text{ and } E(MS(AB)) = \sigma^2 + m\sigma_{ab}^2.$$

2. Subtracting: $E(MSA) - E(MS(AB)) = qm\sigma_a^2$.

3. Equating observed to expected mean squares (ANOVA method of moments) gives

$$\hat{\sigma}_a^2 = \frac{MSA - MS(AB)}{qm}.$$

4. Hence option (a).

Q3.

Consider the random effect model for a two-way classified data with m observations per cell. Which of the following statements is/are correct?

- I. Mean sum of squares due to error is used as error variance for computing the test statistics for $H_{0(A \times B)}: \sigma_{ab}^2 = 0$.
- II. Mean sum of squares due to error is used as error variance for computing the test statistics for $H_{0A}: \sigma_a^2 = 0$.
- III. Mean sum of squares due to error is used as error variance for computing the test statistics for $H_{0B}: \sigma_b^2 = 0$.

A. ☒

B. ☐ II only

C. ☐ III only

D. ☐ II and III

Correct answer: A

EXAM SHORTCUT AI-derived explanation

In a two-way random-effects model only the interaction is tested against MSE; both main effects are tested against MS(AB). So statement I alone is correct.

TIPS & TRICKS

- EMS table: $E(MSE) = \sigma^2$, $E(MS_{AB}) = \sigma^2 + m\sigma_{ab}^2$, $E(MSA) = \sigma^2 + m\sigma_{ab}^2 + qm\sigma_a^2$,
 $E(MSB) = \sigma^2 + m\sigma_{ab}^2 + pm\sigma_b^2$.
- Rule: the denominator is the mean square whose expectation equals the numerator's expectation under H_0 . For H_{0A} that is MS_{AB} , not MSE.
- In the FIXED-effects model everything is tested against MSE - this contrast is the most examinable point in the random/mixed-model topic.

STEP-BY-STEP SOLUTION

- Under $H_{0(A \times B)}: \sigma_{ab}^2 = 0$, $E(MS_{AB}) = \sigma^2 = E(MSE)$, so $F = MS_{AB}/MSE$ is correct. Statement I is true.
- Under $H_{0A}: \sigma_a^2 = 0$, $E(MSA) = \sigma^2 + m\sigma_{ab}^2$, which equals $E(MS_{AB})$, not $E(MSE)$. The correct denominator is MS_{AB} , so statement II is false.
- By the same argument H_{0B} must also be tested against MS_{AB} , so statement III is false.
- Only statement I is correct - option (a).

Q4.

In the case RBD with 'b' blocks and 'v' treatments could be considered as a sample from a larger population, the appropriate two way model could be:

- A. Two way fixed effect model of treatments and blocks
- B. Two way random effect model of treatments and blocks**
- C. Two way mixed effect model of treatments as fixed effect and blocks as random effect
- D. Two way mixed effect model of blocks as fixed effect and treatments as random effect

Correct answer: B

EXAM SHORTCUT[AI-derived explanation](#)

If BOTH blocks and treatments are samples from larger populations, both effects are random, so it is a two-way random effects model.

TIPS & TRICKS

- Fixed: the levels used are the only ones of interest. Random: the levels are a sample from a population of levels.
- 'Could be considered as a sample from a larger population' is the standard verbal signal for a random effect.
- Mixed model applies only when one factor is fixed and the other random - here the stem makes both random.

STEP-BY-STEP SOLUTION

1. In a random effects model the levels actually used are regarded as a random sample from a larger population of possible levels, and inference is about the variance components.
2. The stem states that both the b blocks and the v treatments are such samples.
3. Therefore both block and treatment effects are random, giving a two-way random effects model.
4. Hence option (b).